Optimizing Awareness Campaigns on Social Media for Food Product Recalls Using the Pólya Process

MTHE 493 Thesis

Queen's University

Brooke Thomson (20272357)
Douglass Jeffrey (20294129)
Isabelle Hill (20290600)
Julia Perlin (20268409)
Owen Wade (20270566)

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Abstract

Food recall awareness is critical for protecting public health and reducing exposure to foodborne risks, yet many consumers remain unaware of recall alerts in a timely manner. In Canada, increasing recall frequency and the limitations of current communication strategies highlight the need for more effective, model-driven approaches to information dissemination. This study addresses this challenge by modeling the spread of food recall awareness in digital social networks using a Pólya urn framework on Barabási-Albert graphs.

Two intervention strategies are investigated: a switch network based intervention and direct injection intervention. These methods are evaluated in terms of diffusion speed, cost efficiency, and final network awareness. Consensus is defined as 80% of users being aware of the recall, representing a threshold beyond which the majority of individuals are considered informed and no longer at risk.

Simulation results show that the switch network intervention consistently outperforms direct injection strategies, achieving faster convergence and higher cost efficiency. Under the consensus criterion, the switch network approach reaches the target awareness level at an approximate cost of $6,000 per recall, compared to approximately $8,000 for direct injection.

These results indicate that switch network interventions offer a more cost effective and faster mechanism for achieving widespread awareness compared to direct injection strategies under the proposed model of information spread. This improvement underscores the role of network topology and reinforcement dynamics in enhancing the efficiency of information propagation within public health communication strategies. Future work may extend this framework to dynamic and more heterogeneous network structures, as well as further investigate the sensitivity of outcomes to key parameters such as reinforcement strength, decay rate, and intervention intensity.

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1 Introduction

In this thesis, an analysis of digital advertising strategies for food recall campaigns is explored using the Pólya process framework. The aim is to analyze how information about food recalls propagates through networks and to determine strategies that maximize public awareness while operating within limited budget constraints.

1.1 Objective

This project was designed to model and optimize the dissemination of food recall information using a Pólya network model. The primary goal was to maximize consumer awareness within a fixed time frame while adhering to a predefined advertising budget. To examine effective strategies, two intervention methods are incorporated:

1. Creation of a switch network featuring highly "red-biased" nodes connected to a fixed number of existing nodes.

2. Targeted injection of red balls into nodes with the highest degree, prioritizing the most influential users.

By simulating these interventions, the model identifies the combination of strategies that most efficiently spreads recall alerts throughout the network, providing a quantitative basis for optimizing digital outreach campaigns.

1.2 Motivation

Food recall awareness is critical to prevent adverse health outcomes, including foodborne illnesses, hospitalizations, and, in severe cases, fatalities [1]. Delays in consumer awareness can result in continued consumption of contaminated or mislabeled products, increasing the risk of widespread exposure. According to the Public Health Agency of Canada, foodborne pathogens are responsible for an estimated 4 million illnesses and 3,800 hospitalizations annually in Canada [2]. These public health risks contribute to approximately 150-250 food recalls each year, highlighting a persistent challenge for regulatory agencies [3].

With improvements in food safety systems and enhanced detection methods, the frequency and visibility of food recalls in Canada have increased significantly over the past decade. According to the Canadian Food Inspection Agency (CFIA), the number of food recalls rose from 56 in 2013 to over 130 in 2023, representing an increase of more than 150% [3]. Effective recall communication strategies are essential to reduce these health risks, protect vulnerable populations, and maintain public trust in the safety of the national food supply.

To address these challenges, the Government of Canada has increasingly leveraged digital advertising channels, including social media, programmatic digital ads, and online platforms, to communicate public health and safety information [4]. Among these platforms, Instagram is one of the most widely used and influential, making it a relevant channel for disseminating time sensitive recall alerts [5]. Instagram's high level of user engagement and network based structure make it suitable for analyzing how information spreads online [5]

However, a key challenge in using Instagram for this purpose is the presence of engagement driven feedback loops. User interactions such as likes, shares, and saves influence the platform's algorithm, increasing the visibility of content that performs well early on [5]. This creates a reinforcement effect, where initial engagement can lead to disproportionately large increases in reach. As a result, the spread of information becomes nonlinear and difficult to predict, complicating the design of effective advertising strategies. This type of behavior aligns naturally with the Pólya process framework, where outcomes depend on previous states and reinforcement plays a central role.

At the same time, digital advertising efforts are constrained by limited budgets. During the 2023-2024 fiscal year, the Canadian Food Inspection Agency allocated $1,487,962 to marketing and advertising [6] [7]. Of this

total, approximately $8,400 is allocated per Class I recall, underscoring the importance of highly efficient and targeted dissemination strategies. A detailed breakdown of these expenditures is provided in Section 12.5 of the Appendix [6] [7].

These financial and structural challenges emphasize the importance of optimizing how recall information spreads through networks. By applying a Pólya network approach and comparing different intervention strategies, this project provides insights for decision makers on the optimal allocation of limited marketing resources. Improving recall awareness not only protects public health but also supports trust in food safety institutions and reduces the potential harm caused by contaminated or mislabeled food products.

2 Technical Background

This section provides a technical description of core mathematical models used in the project. It begins with the classic Pólya process and explores how its theoretical framework can be extended to capture the dynamics of contagion spreading across a network. The discussion then turns to switch graphs, outlining the methodologies used in their construction and evolution, particularly when derived from existing graph structures. Further, the section examines the architecture of Barabási-Albert graphs as a means of modeling social media networks, highlighting their relevance in representing real-world connectivity patterns. Finally, it addresses key strategies for mitigating contagion spread, with a focus on optimizing initialization parameters to improve control and influence outcomes within networked systems.

2.1 Pólya Models

2.1.1 Classic Pólya Process

The classic Pólya process is a stochastic reinforcement process and was first introduced by G. Pólya and F. Eggenberger in 1923 [8]. To understand the Pólya process, consider an urn containing R reinforcing red balls and B reinforcing black balls, totaling $T = R + B$ balls ($R, B \in \mathbb{Z}_+$). At time n, a ball is picked from the urn ($n \in \mathbb{Z}_+$). The binary sequence $\{Z_n\}$ denotes the outcomes of the draws:

$$Z_n = \begin{cases} 1 & \text{if the nth chosen ball is red} \\ 0 & \text{if the nth chosen ball is black.} \end{cases} \tag{1}$$

After a red ball is drawn, it is returned to the urn, along with Δ_R balls of the drawn colour ($\Delta_R \in \mathbb{Z}_+$). Similarly, if a black ball is drawn, it is returned to the urn, along with Δ_B balls of the drawn colour ($\Delta_B \in \mathbb{Z}_+$). The values of Δ_R and Δ_B can be equal ($\Delta_R = \Delta_B = \Delta$), as well as time varying or a constant value. Figure 1 provides an iteration of the process, where $R = 5$ and $B = 6$ initially, a red ball selected at $n = 1$, and $\Delta_R = \Delta_B = \Delta = 2$.

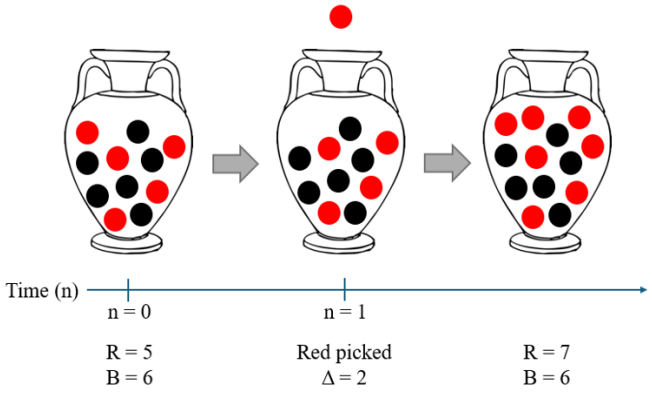


Figure 1: Example of a classic Pólya urn process.

The Pólya process has infinite memory, meaning that once a reinforcing ball is added to the urn, it remains there permanently. However, a finite-memory variation can also be considered [9]. In this version, each newly added ball stays in the urn for only the next M draws ($M \in \mathbb{Z}_+$), after which it is automatically removed. As a result, after the first M draws, the total number of balls in the urn stabilizes and remains constant at $T + M\Delta$. This framework can be used to model contagion dynamics: red balls may represent an "infected" individual and black balls can represent a "healthy" individual.

2.1.2 Network Pólya Contagion Process

The classical Pólya process has been applied to epidemics to model contagion spread through a network [10] [11]. The network Pólya process considers an urn placed at each node of a network. The network of urns can be modeled as an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ is the set of nodes (or vertices) of size $|\mathcal{V}| = N \geq 1$, and $\mathcal{E}$ is the set of edges ($\mathcal{E} \subset \mathcal{V} \times \mathcal{V}$). Let $\{U_{i,n}\}$ denote the urn associated to node i ($i = 1, \ldots, N$) at time n and $\mathcal{N}_i$ the set of nodes (including i) that neighbour node i, ($\mathcal{N}_i = V \in \mathcal{V} : (i, V) \in \mathcal{E}$). Additionally, let R_i, B_i, denote the number of red and black balls at node i, respectively (R_i, $B_i \, \epsilon \, \mathbb{Z}_+$).

At time n, the balls in the urn at each node are combined with its neighbours to form a "super urn". Therefore, the initial number of red and black balls in node i's super urn are $\tilde{R} = \sum_{j \epsilon \mathcal{N}_i} R_j$ and $\tilde{B} = \sum_{j \epsilon \mathcal{N}_i} B_j$. At each node i, a ball is selected from the super urn and Δ_i balls of the colour drawn are added to node i's urn ($\Delta_i \in \mathbb{Z}_+$). The sequence $\{Z_{i,n}\}$ denotes the outcomes of the nth draw at node i:

$$Z_{i,n} = \begin{cases} 1 & \text{if the nth ball chosen for node i is red} \\ 0 & \text{if the nth ball chosen for node i is black.} \end{cases} \tag{2}$$

Figure 2 shows an iteration of the process, with $\{U_{i,n} : i \, \epsilon \, [1, 3]\}$, each with unique initial quantities of red and black balls. A black ball selected at $n = 1$ and $\Delta_i = \Delta = 2$.

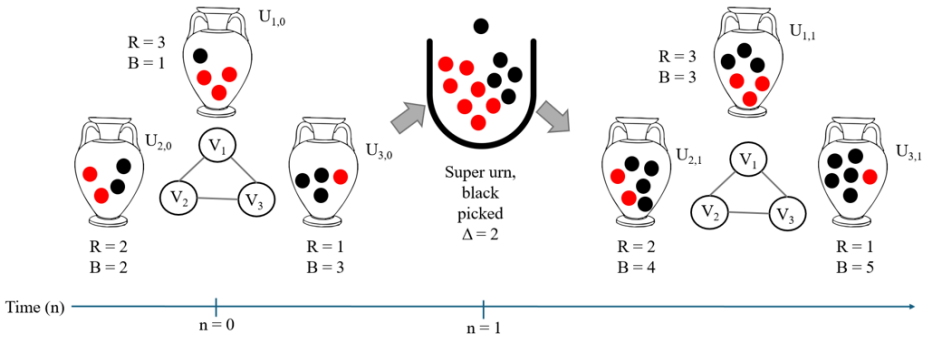


Figure 2: Example of a network Pólya process.

Similarly to the classical Pólya process, a finite memory mode can be enforced in the network Pólya process. In finite memory mode, reinforcing balls added at time n will stay in urn $U_{i,n}$ for M additional cycles ($M \in \mathbb{Z}_+$) and will be automatically removed from the urn at time $n + M + 1$.

2.2 Network Types

2.2.1 Switch Networks

A switch (or threshold) graph is a graph that is constructed by repeatedly adding dominating or isolated nodes to a network [12]. The switch graph network methodology can be applied to existing graphs or used to generate new graphs. New nodes are added probabilistically, or using the outcome of an independent Pólya process. In the case where an independent Pólya process is used, if a red ball is picked at time n, a dominating node V_n, one that connects to all other nodes in the graph $\{V_j, j < n\}$, is inserted into the graph. If a black ball is picked at time n, an isolated node V_n, one that is left unconnected to all other nodes in the graph, is inserted. Figure 3 provides an example of a switch network.

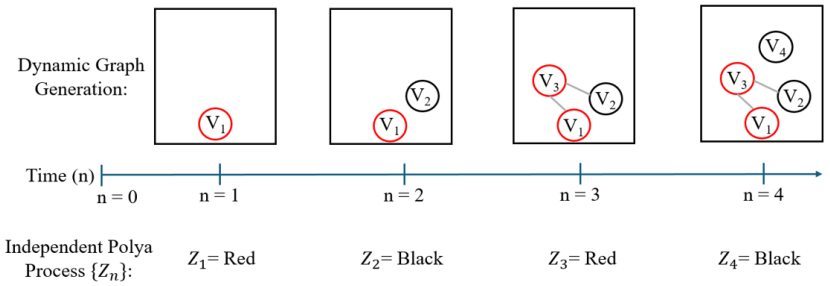


Figure 3: Example of switch network generation from a single urn Pólya process, with dominating nodes represented in red and isolating nodes represented in black.

2.2.2 Barabási-Albert Algorithm

The Barabási-Albert Algorithm is a method for constructing random, scale-free (SF) networks, commonly used to model social networks [13]. The model begins with a small, fixed set of m_0 nodes, typically assumed to be fully connected. The network then grows over time by adding new nodes V_i, each of which forms m connections to existing nodes, where $m \leq m_0$. These connections are assigned probabilistically according to a preferential attachment mechanism: nodes with higher degree are more likely to receive new links. Specifically, the probability of connecting to node i is proportional to its degree, given by the Barabási Centrality Theorem below (Π_i), where k_i denotes the degree of node i at time n, and $\sum_{j=1}^{|\mathcal{V}|-1} k_j$ represents the total degree of the network (equal to twice the number of edges):

$$\Pi_i = \frac{k_i}{\sum_{j=1}^{|\mathcal{V}|-1} k_j} \quad \forall i \in \mathcal{V}, \tag{3}$$

where $|\mathcal{V}|$ is the size of the graph at time n. The probabilistic mechanism governing edge formation during network growth reflects the principle of preferential attachment observed in many real-world social networks, where highly connected nodes are more likely to attract new connections. As a result, nodes with large degrees tend to accumulate links more rapidly than less connected nodes over time. This dynamic leads to the emergence of clustered structures within the graph.

An example of an Albert-Barabási algorithm, with $m_0 = 4$ initial nodes, and a connection rate of $m = 2$ is included in Figure 4.

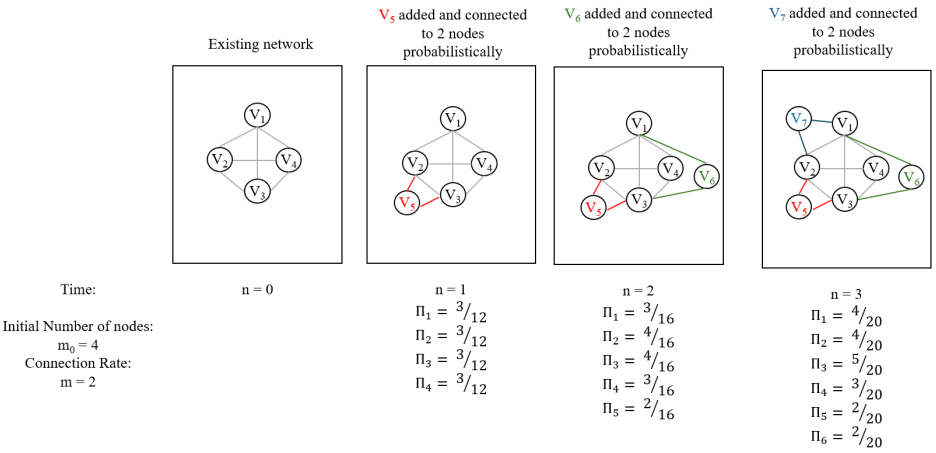


Figure 4: Example of an Barabási-Albert algorithm process.

2.3 Control of Contagion Propagation

2.3.1 Degree Centrality

In order to control the rate at which contagion propagates through a network $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, heuristic strategies can be applied when initializing each node's urn resources, $B_{i,0}$ and $R_{i,0}$ [14]. In particular, resource allocation can be guided by structural properties of the network, such as node degree and closeness centrality. The degree of a node is given by the size of its neighbourhood, $|\mathcal{N}_i|$. Centrality refers to the importance or influence of a node within a network based on its position and connectivity. Its closeness centrality C_i measures how efficiently it can reach other nodes. The closeness centrality measure for node i is defined as:

$$C_i = \frac{1}{\sum_{j \in V} d(i,j)}, \tag{4}$$

where $d(i,j)$ denotes the length of the shortest path between nodes i and j [14].

An alternative objective is to minimize the number of nodes targeted during initialization while ensuring that every node has access to resources within its immediate neighborhood $\mathcal{N}_i$. This approach seeks an efficient placement of resources such that all nodes are "covered" by at least one neighboring node. Two algorithms can be employed for this purpose: one designed for sparse networks (Algorithm 1) and another for dense networks (Algorithm 2) [14]. Both algorithms construct a target set $\mathcal{V}'$ by distinguishing between outer and inner nodes.

Algorithm 1 proceeds by identifying and targeting inner nodes that are adjacent to outer nodes. An outer node $V \in \tilde{\mathcal{V}}$ is defined as one whose neighbourhood is strictly contained within the neighbourhood of another node, $\tilde{\mathcal{V}} = \{i \in \mathcal{V} \mid \exists\ j \in \mathcal{V}\ s.t\ \mathcal{N}_i \subset \mathcal{N}_j\}$. In contrast, an inner node $V \in \mathcal{V}'$ is one whose neighbourhood $\mathcal{N}_i$ is not a strict subset of any other node's neighbourhood: i.e., $\mathcal{V}' = \{i \in \mathcal{V} \mid \mathcal{N}_i \not\subset \mathcal{N}_j\ \forall j \in \mathcal{V}\}$. These inner nodes are added to the target set $\mathcal{V}'$, after which all nodes in their neighbourhoods are removed from further consideration. The procedure is then repeated on the reduced graph. A detailed description of Algorithm 1 is provided in Figure 5 below [14].

$V \leftarrow$ set of all nodes
$V' \leftarrow$ target set
$V_{test} \leftarrow$ nodes to test for this loop
Start with $V' = \emptyset$, $V_{test} = V$
while $V_{test} \neq \emptyset$ **do**
  Identify outer nodes:
  Set $V_{outer} = \{i \in V_{test} | \mathcal{N}_i' \subset \mathcal{N}_j',\ \text{for some } j \in V_{test}\}$
  if $V_{outer} = \emptyset$ **then**
    $V' = V' \cup V_{test}$
    break
  end if
  Target nodes adjacent to outer nodes:
  Set $V_{inner} = V_{test} \backslash V_{outer}$
  Set $V_{added_targets} = \{i \in V_{inner} | i \in \mathcal{N}_j' \text{ for some } j \in V_{outer}\}$
  Set $V' = V' \cup V_{added_targets}$
  Update parameters for next loop:
  Set $V_{coverage} = \{i \in V_{test} | i \in \mathcal{N}_j' \text{ for some } j \in V_{added_targets}\}$
  Set $V_{test} = V_{test} \backslash V_{coverage}$
end while

Figure 5: Algorithm 1 for determining optimal target set $\mathcal{V}'$ for initialization.

Algorithm 2 accounts for the fact that dense networks may not contain nodes whose neighbourhoods are strict subsets of those of other nodes. To address this, it introduces a preliminary step in which all nodes are ranked according to their closeness centrality C_i. Nodes are then added sequentially to the target set $\mathcal{V}'$ in this order until the union of their neighbourhoods covers the entire network. A detailed description of Algorithm 2 is provided in Figure 6 below [14].

$V \leftarrow$ set of all nodes
$V' \leftarrow$ nodes to be targeted
$C_i \leftarrow$ centrality score of given node
Without loss of generality, assume $C_1 > C_2 > \cdots > C_N$, where N is the network size
Start with $V' = \emptyset$, $V_{coverage} = \emptyset$, n=0
while $V_{coverage} \neq V$ **do**
  Add most central node to target set:
  $V' = V' \cup \{n\}$
  Update loop parameters:
  Set $V_{coverage} = \{i \in V | i \in \mathcal{N}_j',\ \text{for some } j \in V'\}$
  n++
end while
Optional:
for $n = |V'| : 1$ **do**
  Let $V_{coverage} = \{i \in V | i \in \mathcal{N}_j',\ \text{for some } j \in V' \backslash \{n\}\}$
  if $V_{coverage} = V$ **then**
    Set $V' = V' \backslash \{n\}$
  end if
end for

Figure 6: Algorithm 2 for determining optimal target set $\mathcal{V}'$ for initialization.

Algorithms 1 and 2 can be applied to a network to identify an optimal set of target nodes, $\mathcal{V}'$. Once

$\mathcal{V}'$ is determined, initialization parameters can be allocated using different schemes. Rather than assigning resources uniformly across nodes, improved performance can be achieved by distributing them proportionately based on a combination of node degree and closeness centrality. In particular, for each node $i \in \mathcal{V}'$, resources are allocated according to: $\frac{|\mathcal{N}_i|C_i}{\sum_{j \in V} |\mathcal{N}_i|C_i}$ if $i \in \mathcal{V}'$, and set to zero otherwise [14].

2.3.2 Eigenvector Centrality

Eigenvector centrality varies from degree centrality in that it accounts for the entire structure of the network when selecting targets for intervention, rather than simply selecting those with many direct connections. Formally, eigenvector centrality assigns a score to each node proportional to the sum of its neighbours' centrality scores. The relationship is defined by the following eigenvector equation:

$$A\mathbf{x} = \lambda\mathbf{x} \tag{5}$$

where A is the adjacency matrix representing the network, λ is A's largest eigenvalue, and $\mathbf{x}$ is the eigenvector centrality score.

3 Problem Definition

Food recalls pose significant public health risks, yet many consumers remain unaware of recall alerts in a timely manner. Despite the increased frequency of recalls in Canada and government investments in digital marketing, there is limited understanding of how different communication strategies influence the speed and reach of recall information. Regulatory agencies face the challenge of distributing information efficiently under budgetary constraints while ensuring that the majority of consumers are reached before potentially hazardous products are consumed. Additionally, the stochastic nature of digital networks and consumer engagement patterns complicates the prediction of campaign effectiveness. These challenges highlight the need for a systematic, model-based approach, such as a Pólya network, to optimize the allocation of limited resources and improve campaign outcomes.

3.1 Problem Statement

This research seeks to identify and evaluate the most effective digital advertising strategies for food recall campaigns to maximize consumer awareness within a fixed period and budget. The study focuses on understanding how information spreads through a social network and how different intervention methods, specifically switch networks with red-biased nodes and targeted injection of red balls into high-degree nodes, impact the reach and speed of recall alerts. By modeling these dynamics, the research aims to provide guidance for allocating limited advertising resources efficiently, ensuring that recall information reaches the largest possible portion of the population in a timely manner and ultimately reducing the public health risks associated with delayed consumer awareness.

4 Design Process

The objective of the project is to optimize the use of social media for the spread of public awareness on social issues, specifically regarding food safety. Two strategies will be tested to understand the best approach to maximize awareness under cost constraints. Each strategy will be tested under numerous iterations with varying parameters to ensure an accurate representation and conclusive results.

4.1 Network Design

To accurately model the spread of food recall awareness over social media networks, the underlying network structure was required to reflect real-world connectivity patterns. The social platform selected for this analysis was Instagram, as it is among the most widely used applications in Canada [15]. The network was modeled using the Barabási-Albert model, which characterizes systems that evolve through growth and preferential attachment.

In the classic Barabási-Albert model, networks grow through the continuous addition of new nodes, which preferentially attach to existing nodes with higher degrees. In other words, the probability of forming a connection is proportional to a node's current level of connectivity. This process gives rise to a scale-free degree distribution, where a small number of nodes accumulate disproportionately many connections compared to the rest of the network. Applied to social media, nodes represent individual users and edges correspond to pathways through which information flows. Preferential attachment reflects the tendency of users to engage with and share content from highly connected or authoritative sources, such as government agencies, news organizations, or influencers. As a result, these high-degree nodes function as influential hubs that disproportionately influence the spread of information across the network.

This network structure implies that awareness propagation is non-uniform, with a small subset of nodes playing a critical role in reaching a large portion of the population. As such, the Barabási-Albert model was determined to be an appropriate representation of social media networks for evaluating food recall awareness strategies. Multiple different Barabási-Albert networks were generated and used for testing. Parameters such as number of connections per node, as well as number of nodes in the networks were varied.

Each strategy was evaluated on three base Barabási-Albert networks, each initialized with $m_0 = 64$ nodes. Due to uncertainty in the connectivity of real-world social networks, the connection rate during network growth was varied. The three trials were conducted for values of $m = 1, 3$ and 5. Each network was grown to a size of ten million nodes, and used for each simulation iteration, allowing for analysis of the model's sensitivity to network density and structure. The network generated for the $m = 1$ rate is shown in Figure 7. Scaling the base network by a factor of 2.1 can approximate the network of Instagram users in Canada [15]. The Pólya urn process was then applied to this topology to simulate the stochastic dynamics of awareness propagation across the network.

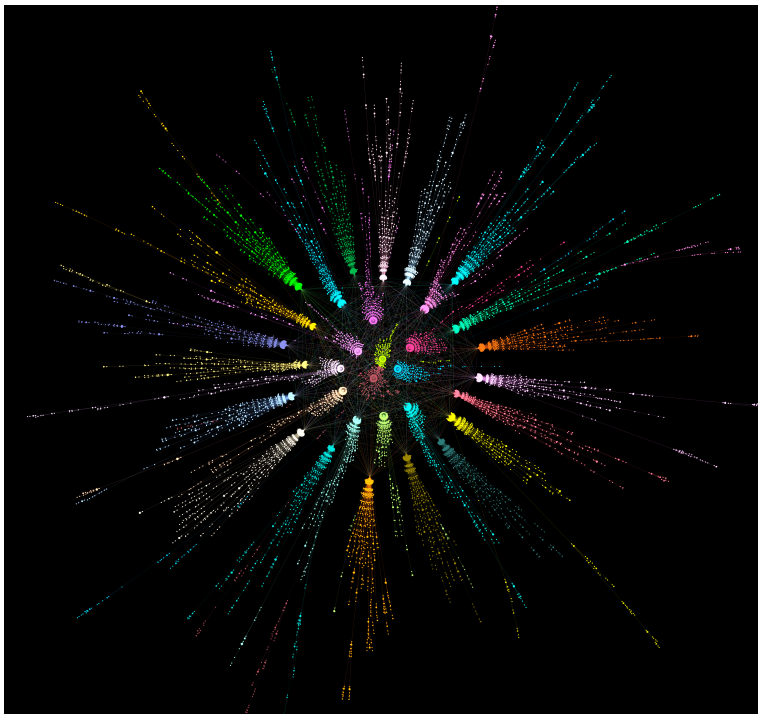


Figure 7: The Barabási-Albert network of size 10,000 generated for $m = 1$.

4.2 Applying the Pólya Process

Each node was initialized to be unbiased, consisting of one red ball and one black ball. In the context of awareness spread on social media, the red balls represent "aware" and the black balls represent "unaware". "Aware" represents receiving information regarding the food recall and its associated risks, making them more likely to take precautionary actions or share this information with others. Conversely, "unaware" represents not being exposed to recall information, reflecting a lack of risk perception and a continued likelihood of consuming the recalled product.

To capture the effects of spreading information on social media on a broader scale, the network Pólya process was applied. Taking each node in the network graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ as an individual user on social media, the network Pólya process demonstrates the impact of the campaign on the public. The balls inside each node of a neighbourhood are combined to form a super urn. By applying the Pólya process to the super urn, the neighbourhood develops a bias toward a particular ball colour. This bias reflects the influence of public awareness on an individual's likelihood of becoming aware of a recall.

A finite-memory network Pólya process was implemented for all simulations. At each timestep n, the reinforcement parameters were equal and constant across all nodes, $\Delta = \Delta_R = \Delta_B = 1$. This setup implies that following each draw $Z_{i,n}$, a single ball of the observed colour is added to the corresponding urn $U_{i,n}$. To reflect that users gradually forget or deprioritize food recall information over time, a finite memory of $M = 72$ timesteps was used. Each reinforcing ball introduced at time n remained in the urn for M subsequent draws and was automatically removed at time $n + M + 1$. Given that each timestep represents one hour, this corresponds to a memory of three days [16]. This finite memory was applied consistently across all models and simulation cases.

4.3 Determining Model Consensus

Awareness in the network was quantified using a consensus metric, defined as the proportion of red balls across all nodes at the end of the simulation. Denoting R_i and B_i as the number of red and black balls in node i,

for all $i = 1, \ldots, N$, where N is the number of nodes in the network, the overall network consensus is given by:

$$\mathcal{C} = \frac{\sum_{i=1}^{N} R_i}{\sum_{i=1}^{N} (R_i + B_i)} \tag{6}$$

In the application of health and safety, $\mathcal{C} \geq 0.8$ is defined as a reasonable threshold to assume consensus [17]. This criterion can be applied to food recall scenarios, where achieving this level of awareness suggests sufficient dissemination of critical information to effectively reduce public health risk. Given that both contexts rely on widespread information diffusion to influence individual behaviour, it is reasonable to adopt the same threshold. Thus, in our simulations, consensus is defined as being achieved when at least 80% of users are aware of the food recall and are likely no longer at risk.

4.4 Intervention Strategies

Two methods were evaluated to determine the most effective way to reach consensus. The first strategy models interventions as new, dominating nodes, containing a high proportion of red balls being inserted via the switch graph methodology. The second strategy directly injects red balls into a fixed quantity of nodes in the graph, chosen by their centrality.

4.4.1 Model 1: Intervention Using a Switch Network

The first intervention strategy adds new nodes to the base network according to the switch graph methodology. The model begins with a pre-existing graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ of size $|\mathcal{V}|$ =10,000,000, with each node's urn containing some distribution of black and red balls. At each timestep $n = 1, 2, ...$, the network Pólya process is run. An independent Pólya process Z_p is run to add new nodes into $\mathcal{G}$ according to the switch graph methodology, with timestep $p \leq n$. At time p, if a red ball is picked, a dominating node is inserted into $\mathcal{G}$ and connected to all (or a fixed percentage) of nodes in the network. If a black ball is picked, an isolated node is inserted into $\mathcal{G}$.

In this strategy, awareness is introduced into the network through highly influential sources. The insertion of dominating nodes with a high concentration of red balls reflects real-world entities such as news outlets, government agencies, or highly connected individuals defined as influencers. The high concentration of awareness in these entities are used in the Pólya Process to represent actively sharing food recall awareness. Due to their high connectivity, these nodes significantly alter the composition of local super urns, increasing the likelihood that neighbouring nodes transition from unaware to aware states. As a result, the model captures how centralized, high-impact communication can rapidly accelerate the spread of awareness across large portions of the network.

To evaluate the effectiveness of the intervention strategy, the number of red balls within each intervention node's urn, the number of intervention nodes introduced at each intervention step of the Pólya process, and the probability of an intervention occurring were systematically varied. In the context of food recall awareness, these parameters correspond to the intensity of information dissemination and user reception from a source, the scale of simultaneous outreach efforts, and the frequency of intervention events. The influence of these parameters on awareness propagation is detailed in Section 6.1.

4.4.2 Model 1 Assumptions

The model has the following assumptions:

- The base Barabási-Albert network remains static during the simulation. No additional nodes or edges are added through natural network growth, outside of the intervention strategy.
- Each simulation runs for 720 time-steps. Each time-step represents one hour; therefore, the simulation represents approximately one month.

- In each simulation, all added nodes have identical urn composition, meaning that there is no variation in a user's behavioural characteristics.
- Intervention nodes have immediate and uniform influence over all connected nodes upon insertion.
- Information transfer occurs only through the defined network edges. External sources are not considered.
- Recall information shared is accurate and uniformly interpreted by users.

4.4.3 Model 2: Targeted Injection of Red Balls into Optimal Nodes

The second intervention strategy involves directly injecting red balls into selected nodes within the existing network. The pre-existing graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is of size $|\mathcal{V}|$ =10,000,000, with each node's urn containing some distribution of black and red balls. At each timestep $n = 1, 2, ...,$ the network Pólya process is run. At a fixed timestep p ($p \leq n$) the red balls are injected directly into a fixed number of node's urns. Two methods were used to determine which nodes were selected for intervention. The first method uses degree centrality, selecting nodes with the highest number of direct connections. The second method uses eigenvector centrality, which prioritizes nodes that are not only highly connected, but also connected to other highly influential nodes.

Awareness is introduced through targeted injection of red balls into selected nodes, representing food recall messages or advertisements delivered directly to highly connected users. Nodes selected via centrality measures correspond to individuals with significant influence over information dissemination, either through direct reach (degree centrality) or connectivity to other influential nodes (eigenvector centrality). Increasing the proportion of red balls in these nodes, increases the likelihood of awareness propagation through their local neighbourhoods. This intervention strategy leverages key individuals to efficiently propagate food recall awareness throughout the network.

To optimize the intervention strategy, several parameters were systematically varied. The number of red balls injected into each selected node's urn was adjusted to represent the effectiveness or quality, of the initial recall awareness message, with higher values corresponding to greater user receptiveness. Additionally, the number of nodes selected for injection was varied to reflect the initial scale of the outreach effort, where increasing the number of targeted nodes leads to broader initial awareness across the network. The influence of these parameters on awareness propagation is detailed in Section 6.2

4.4.4 Model 2 Assumptions

The model has the following assumptions:

- The base Barabási-Albert network remains static during the simulation. No additional nodes or edges are added.
- Prior to injection, the model is run for 144 initial time-steps to allow finite memory to initialize correctly.
- All selected nodes are injected on the same timestep, with the same number of red balls. This assumption implies uniform message quality across targeted users.
- Injected red balls only affect the internal state of the selected nodes and do not alter the underlying network structure.
- Each simulation runs for 720 time-steps. Each time-step represents one hour; therefore, the simulation represents approximately one month after injection.
- Information transfer only occurs through defined network edges. External sources are not considered.
- Recall information shared is accurate and uniformly interpreted by users.

4.5 Cost Analysis

To compare these models within the real-world context of disseminating food recall information, it is necessary to model the economic costs associated with each intervention method.

The direct injection intervention represents targeted advertising, in which information is delivered to specific users based on their interactions with content on social media. Corporations such as Meta and X Corp collect and analyze user data to curate content feeds and deliver relevant advertisements. In practice, organizations pay for such advertisements on social media, using a cost per thousand impressions (CPM) pricing model. In Canada, CPM rates typically range from $5 to $15, with an average cost of approximately $8 in 2025 [5] [18]. Based on this information, the simulation cost of the direct injection intervention was modeled conservatively as a function of the number of interventions and per-intervention connections, where $N_{injection}$ represents the total number of injection interventions, and $T_{per\ injection}$ represents the number of injection targets in each intervention.

$$Cost_{sim\ total} = \frac{CPM}{1,000} * N_{injection} * T_{per\ injection} \tag{7}$$

The switch network intervention models the dissemination of information through posts made from highly followed accounts. Government operated social media accounts often have audiences ranging from tens of hundreds to millions of followers [19]. While the reach of such accounts is substantial, the actual spread of information depends on user engagement patterns, including platform usage frequency and network connectivity. Unlike the direct injection intervention method, the cost of a switch network intervention is fixed for each intervention. Once content is published, it can propagate through the network organically at no additional cost. The primary expense is the labour required to create, review, publish, and monitor the post. Although, content creation costs also apply to direct injection interventions, these costs are fixed and equivalent across both methods, and are therefore excluded from the comparative analysis. Under these assumptions, the net cost of producing a post is given by the total time required to draft, review, publish, and monitor the content, multiplied by the hourly wage of the individual performing these tasks.

Estimates place the annual salary of a social media communications specialist at approximately $60,000, corresponding to an hourly wage of $30 [20]. Assuming a conservative estimate of two and a half hours of work per post, the cost per intervention is approximately $75, implying that each switch network intervention incurs this fixed cost. Let $N_{intervention}$ denote the number of switch network interventions. The total simulated cost is then modeled as:

$$Cost_{sim\ total} = 75 * N_{intervention} \tag{8}$$

To ensure comparability across simulations of varying scale, total costs of both interventions were adjusted by a scaling factor S, ranging from 10-100 depending on simulation scale. This scaling reconciles real world costs with simulated environments and preserves meaningful variation in results. Under this framework, in a simulation with 10 million nodes, a single injection targeting 10,000 distinct target nodes corresponds to $8,000 per recall, while the switch network intervention results in a total cost of approximately $6,000 per recall.

Within this framework, the number of balls introduced through each intervention, as well as the number of balls added at each step of the Pólya process, represents the perceived severity or importance of the information. A larger number of introduced balls corresponds to higher informational impact, increasing the likelihood that exposed users will further disseminate the recall information throughout their local network.

4.6 Evaluating Tool Limitations and Trade Offs

Prior to defining the simulations, it is crucial to understand the limitations and trade-offs included in each proposed solution.

The Pólya urn process proposes a methodology for modeling reinforcing behaviour of advertisement campaigns within a network. However, it is limited in accuracy, as the impact of an ad on individuals is difficult to quantify, and thus the value of reinforcing delta, Δ_B, Δ_R is chosen rather arbitrarily. Additionally, while the Barabási-Albert model is a commonly used to model social networks, it provides a simplified approximation of real-world connectivity. Social media networks are highly complex, as both the number and strength of connections vary significantly between users. In this model, a reduced level of connectivity was assumed to reflect the connections where users are likely to share information. While this assumption improves computational efficiency, it does not capture the variability of real information propagation.

Due to these limitations and model assumptions, it is important to consider that although the model may be accurate in modeling the trend of awareness spreading through a network, the quantifiable results may not be entirely accurate. When assessing the final projected costs, a certain margin of error should be considered. Additionally, parameter variations will be run and compared across simulations, reducing their associated uncertainties.

4.7 Evaluation Metrics

Evaluation metrics were developed to analyze and compare the performance of each intervention strategy. Two categories of metrics were considered: engineering metrics and technical metrics. These metrics will be applied to the simulation results to assess model performance and objectively weigh project goals. Table 1 outlines the evaluation metrics and their respective definitions.

Metric	**Metric Description**
Economic	Evaluates the economic viability of the model by comparing its projected cost at consensus to the current average cost of food recall campaigns.
Environmental	Assesses the environmental impact associated with implementing the solution. Considerations include materials usage, energy consumption, carbon emissions, and waste products generated through long-term implementation of the model.
Social	Considers the social impacts associated with the implementation of the model. It considers the accessibility and inclusion of all members of the public in receiving information, in the context of food recall campaigns.
Feasibility	Assesses if the implementation of the mathematical model is feasible. It considers the real-world infrastructure of advertisement platforms and the ability to distribute advertisements in a targeted manner.
Safety	Assesses the risk to public health and safety associated with implementing the model, for food recall campaigns.
Complexity	Evaluates the complexity of the model with respect to physical hardware requirements, including storage capacity, simulation time, and computational resource usage.
Scalability	Assesses the model's scalability to larger populations and increasingly complex social media networks, including the impact of scaling on consensus trends.
Ability to reach consensus	Assesses the ability of the model to reach a minimum of $\mathcal{C}\%$ within the simulation time.
Propagation speed	Assesses the speed at which $\mathcal{C}\%$ is achieved in the network.

Table 1: Summary of engineering and technical metrics used to evaluate model performance based on simulation results.

Each metric was assigned a weighting from one to five. A score of one denotes the lowest importance and relevancy level to the project, while a score of five denotes the highest, with intermediate scores (two to four) representing varying decrees between these bounds. This weighted evaluation matrix (WEM) is shown in Table 2. In Section 7, the WEM is then completed for each model by assigning a score from one to five for each metric.

Metric	**Weighting (1-5)**
Economic	4
Environmental	2
Social	3
Feasibility	3
Safety	5
Complexity	3
Scalability	2
Ability to reach consensus	5
Propagation speed	3

Table 2: Weightings assigned for each evaluation metric.

Public safety and consensus achievement were prioritized above all other metrics. Given the severity of food recall campaigns, maximizing population-wide awareness is essential. Propagation speed and economic impact were also assigned high importance, each receiving a weighting of four. Timely dissemination of information is critical, and given fixed government funding to recall initatives, the proposed model must remain within budget to ensure feasibility.

Social, feasibility, and complexity metrics were assigned a moderate weighting of three. While these factors remain important, they were considered less critical than those with higher weightings. Ensuring an equitable and inclusive distribution of awareness across the network is essential so marginalized groups are not disproportionately uninformed about food recalls. Feasibility is also a key consideration, as the proposed models must extend beyond simulation and be implementable in practice, particularly through integration with existing social media and advertising frameworks. Additionally, the complexity of each strategy can influence its performance and ease of implementation.

Environmental impact and scalability were each assigned a weighting of two. Although the long-term considerations such as material usage, energy consumption, and emissions remain important, the proposed models rely on integrating advertisment strategies into existing media infrastructures, thereby reducing their relative environmental impact compared to other factors.

Further justification of the scores assigned to each metric can be seen in Table 6 in the Appendix.

5 Network Design Computational Optimizations

Throughout the design process, the simulation required balancing trade-offs between memory usage, computational efficiency, accuracy, and execution time. This section outlines the key decisions and optimizations made at each stage. The code implementation supporting these simulations is available on GitHub; refer to Section 12.1 in the Appendix for the repository link.

5.1 Iterative Design Process

The initial implementation of the simulation was developed entirely in Python, with each node and its associated urn represented as a discrete object in memory. The simulation pipeline consisted of three main stages: instantiating all node and urn objects, generating edges to construct a Barabási–Albert graph, and executing the Pólya process over the resulting network. This design proved to be poorly suited for large-scale simulations. Since the primary objective of the model was to represent a social network comprising of millions

of users and to study the dynamics of information propagation, the object-oriented implementation did not scale effectively to networks of this size.

The most significant limitation was execution time. Constructing the super urn for each node required sequential access to all neighbouring nodes via function calls, often involving multiple calls per neighbour. For a graph with 1,000 nodes and an average degree of six, a single iteration of the Pólya process required 12,000 additions to construct the super urns, 6,000 random draws to select balls, another 6,000 additions to redistribute the balls back into the urns, and the cleanup of 1,000 temporary super urn objects. As this computational cost was incurred at every step, the simulation became prohibitively slow even at relatively modest scales.

Memory usage also posed a constraint. Representing each node and urn as separate objects introduced substantial overhead, limiting the size of the network graph to approximately 1,000 nodes. At this scale, each node would need to represent an aggregate of many individuals, resulting in a loss of granularity that undermined the realism of the model.

To address these limitations, the simulation was restructured using NumPy, a library designed for efficient numerical computation. In this revised approach, urn states were stored in a single array, where each row corresponded to a node and each column to a ball type. Similarly, the graph's edges were represented using an adjacency list, where each row corresponds to a directed connection between two nodes. The shift from object-oriented to data-oriented design significantly reduced memory overhead and enabled a more compact representation of the system. The execution time was also improved by this shift. Instead of performing many sequential operations, the simulation could now use batched and vectorized array operations over the urn and adjacency matrices. With these improvements, the simulations scaled up to 500,000 nodes. In addition, the optimizations made it possible to incorporate additional features, such as urn memory, into the Pólya process.

Then, graph generation was the main bottleneck. In a Barabási-Albert network, each new node must connect to some number of existing nodes with attachment probabilities proportional to their current degree. Thus, every node addition is dependent on the current state of the graph and must be generated sequentially. As a result, the time required to add each successive node scales linearly as the graph grows, making the generation of very large graphs prohibitively slow.

To fix this bottleneck, the node generation procedure was approximated by adding large batches of nodes simultaneously, where all nodes in a given batch selected their connections based on the same snapshot of the graph. Since the parameter m governing the number of connections each new node forms is typically small, this approximation introduced negligible deviation from the true Barabási-Albert graph and does not alter the Pólya process dynamics.

This batching optimization allowed the simulation to scale to networks with 10 to 15 million users. However, even at this stage, running a single step of the Pólya process on graphs of this scale could take minutes to hours, as simulations contained over one billion edges. NumPy alone was no longer sufficient to handle computations at this scale efficiently.

Two additional optimizations were then implemented to address this bottleneck. First, the edge array was stored as a compressed sparse row (CSR) matrix, which further reduced memory usage and allowed the Pólya process to be reformulated in terms of sparse matrix multiplication. Second, GPU acceleration was added using the `CuPy` library in Python, allowing the simulation to run on Nvidia GPU's. This CPU GPU pair allowed significant speedup by enabling parallel computation of the many operations involved in each Pólya process step.

5.2 Quantifying Performance Improvements

To evaluate the impact of the optimizations described above, the performance of both the initial and final implementations were measured across a range of graph sizes. Figure 8 compares the time required by each approach to generate small graphs and execute 100 steps of the Pólya process. The results show that the compute time of the initial design increases at a greater exponential rate compared to the final implementation, highlighting the performance gains achieved through the revised architecture. For larger graphs on the order of 10^5 to 10^8 nodes, the runtime of the initial design quickly exceeded practical computational limits.

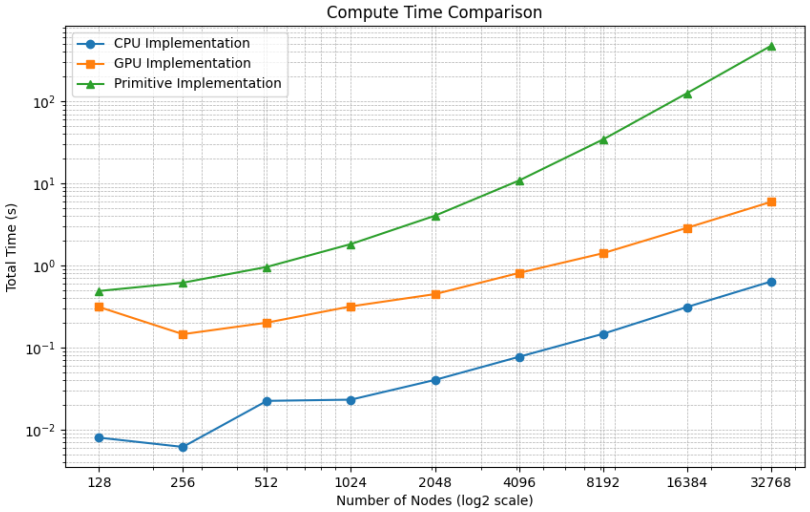


Figure 8: Compute time performance analysis for initial (primitive) simulation design and final design ran on CPU and GPU CPU pair.

The results presented in Figures 9 and 10 demonstrate the performance improvements achieved through GPU-based implementations relative to the CPU baseline. While the CPU runtime increases rapidly with graph size, both GPU implementations exhibit more favourable scaling behaviour at larger scales. In particular, for graphs in the range of 10^6 to 2×10^7 nodes, the local GPU implementation outperforms the CPU-only approach by more than a factor of ten, proving the effectiveness of the optimization.

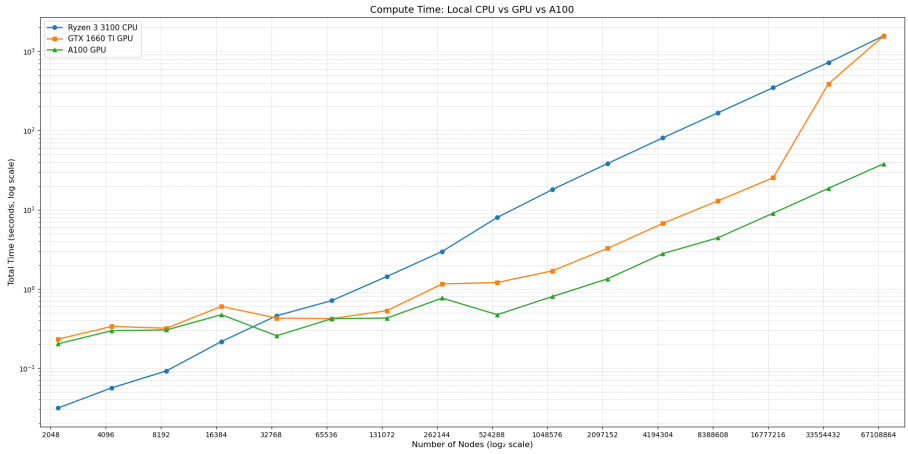


Figure 9: Compute time performance comparison between simulated design on local CPU, local GPU and CPU pair (1660 Ti), and modern data center GPU (A100).

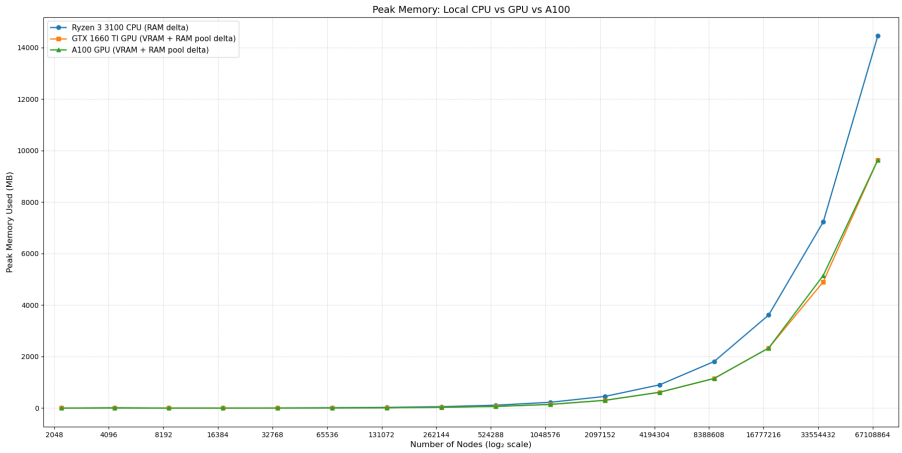


Figure 10: Peak memory usage comparison between simulated design on local CPU, local GPU and CPU pair (1660 Ti), and modern data center GPU (A100).

6 Simulation Results

This section examines the key parameters influencing awareness propagation within the network model. By systematically varying factors such as network connectivity, intervention strength, frequency, and scale, the effects on convergence and overall cost are evaluated. The results reveal trade-offs that inform the development of more efficient intervention strategies. To assess the effectiveness of both intervention approaches, each parameter is varied independently.

6.1 Model 1: Intervention Using a Switch Network

The simulations follow the setup described in Section 4.4.1 and Section 4.4.2, where each user's urn was initialized with one red and one black ball to represent an unbiased awareness state, and the reinforcement parameter, Δ, was set to one. Intervention nodes were connected to 1,000 users, reflecting common social media marketing metrics such as cost per thousand impressions [18]. A memory window of 72 time steps (equivalent to three days) was implemented to model the decay of user awareness over time [16]. To evaluate awareness propagation, the proportion of nodes with at least 80% red balls was recorded at every tenth time step, providing a measure of the percentage of users exhibiting an awareness bias. All simulations in this section use a fixed probability of intervention of 10%, with the number of balls in each intervention urn set to 2,000, the number of intervention nodes per step held constant at five, and each new node in the Barabási-Albert graph connecting to one existing node. The costs for Model 1, the switch network intervention, are calculated as explained in Section 4.5.

6.1.1 Number of Barabási-Albert Connections (m)

The first parameter examined was the number of connections in the Barabási-Albert graph (m), which represents the level of connectivity within the network. In this model, m reflects the number of close contacts through which a user is likely to share information about food recalls. The results of this simulation are shown in Figure 11.

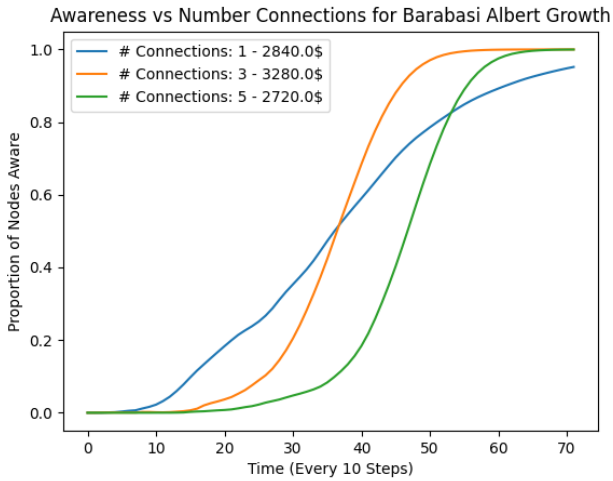


Figure 11: Awareness percentage in model for different number of connections for each new node in Barabási-Albert graph.

Figure 11 highlights that for three and five connections, full consensus is achieved, while for one connection, nearly full consensus is achieved (approximately 90%). The results also show that with fewer connections in the initial graph, convergence begins earlier but occurs at a slower rate. For networks with multiple connections per node, convergence takes longer to begin, but once convergence starts, it occurs much more rapidly. These results indicate that as the number of connections increases, awareness takes longer to begin propagating through the network, but once propagation starts, it spreads more quickly than in networks with fewer connections. Based on these results, networks with more connections may require more intervention nodes at the start of the simulation to encourage earlier propagation, so that users become aware of recalls closer to when they are released. If awareness takes a month and a half to begin spreading, many individuals may continue using recalled items before becoming aware of the recall.

6.1.2 Number of Red Balls Per Intervening Node

The next parameter examined was the number of red balls in each intervention node's urn, which is directly related to the overall cost of the awareness campaign. The results are shown in Figure 12.

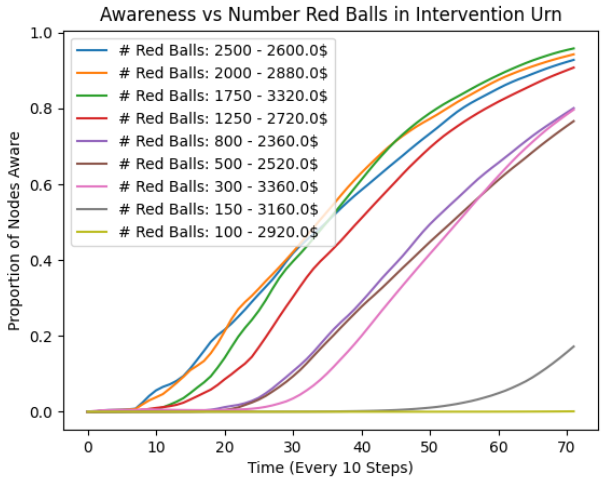


Figure 12: Awareness percentage in model for different amounts of red balls in intervening node urns.

Figure 12 indicates that beyond 1,250 red balls, the improvement in convergence is minimal. The higher cost observed for the urn with 1,750 red balls suggests that significantly more interventions occurred, which explains the slightly higher proportion of aware nodes compared to simulations with fewer red balls. From the results, the case with 1,250 red balls appears to have a moderate cost while achieving performance similar to cases with significantly more balls. As a result, performing additional interventions with approximately 1,250 red balls would likely produce comparable outcomes to those with more red balls, while maintaining a significantly lower cost.

6.1.3 Probability of Intervention

The probability of an intervention step occurring was the next parameter examined, with the tested values shown in Figure 13. This parameter assesses how the number of nodes added at each intervention step influences both the level of consensus achieved and the associated cost, with the goal of determining the minimum number of nodes required for efficient convergence. The results are shown in Figure 13.

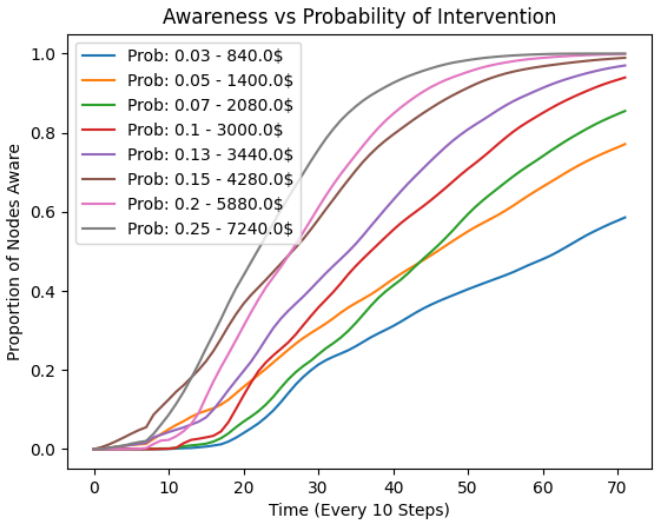


Figure 13: Awareness percentage in model for different intervention step probabilities.

The simulation comparing differences in intervention probability, shown in Figure 13, performs as expected. Higher intervention probabilities lead to earlier and faster convergence to consensus. The results also show that cost increases with probability, since a higher probability results in more frequent interventions and, in turn, a higher overall cost. The results further indicate that for intervention probabilities greater than 10%, the increase in consensus is minimal, while the increase in cost is significant. The main difference between the cases is the time at which the curves begin to converge toward consensus. For recall applications, it is important for convergence to begin as close to the time of the recall as possible so that the public becomes aware and can respond accordingly.

Based on these results, modeling the probability of an intervention step as an exponential decay function would be more effective. An exponential decay approach allows for a high probability of intervention at the beginning of the simulation, which gradually decreases over time as awareness spreads through the network. This approach better reflects real-world scenarios, where the most intensive effort to spread awareness occurs immediately after a recall and decreases over time. In addition, this method reduces overall cost by limiting unnecessary interventions once awareness is already propagating through the network.

6.1.4 Number of Connections Per Intervention Step

The final parameter examined was the number of connections per intervention step, as shown in Figure 14.This parameter assesses how the number of nodes added at each intervention step influences both the level of consensus achieved and the associated cost, with the goal of determining the minimum number of nodes required for efficient convergence. The results are presented in Figure 14.

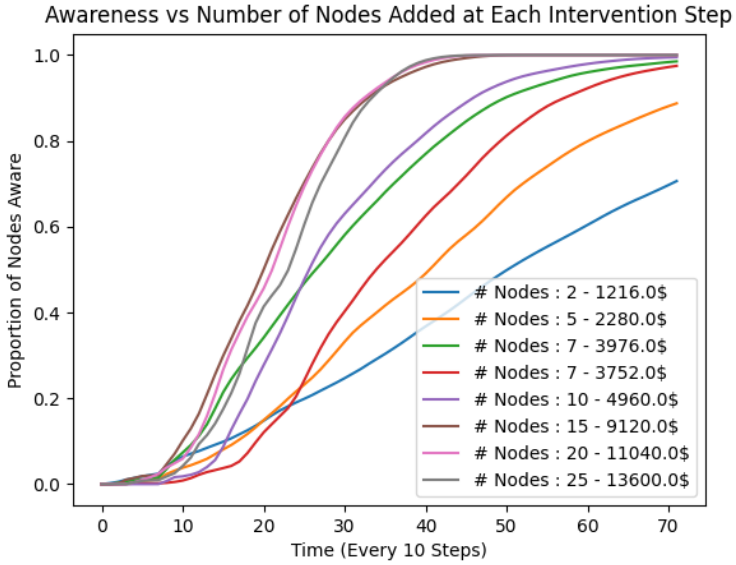


Figure 14: Awareness percentage in model for different number of intervention nodes added at each intervention step.

The simulation comparing the number of intervention nodes added at each intervention step, shown in Figure 14, behaves as expected. Increasing the number of intervention nodes at each step leads to a faster rate of convergence to an aware consensus, but also results in a higher overall cost. The results show that beyond approximately seven intervention nodes, the improvement in convergence is minimal, while the increase in cost remains substantial. The main difference between the cases is the point at which the curves begin to converge toward consensus, with simulations using more nodes reaching this stage earlier and progressing more quickly. These results suggest that using an exponential decay function for the intervention probability may improve efficiency, as it would concentrate more intervention nodes at the beginning of the simulation and fewer toward the end. Further analysis is needed to evaluate the combined effect of varying the number of intervention nodes with an exponentially decaying intervention probability.

6.1.5 Revisiting Probability of Intervention

Based on the previous results, two additional simulations were performed to evaluate the effectiveness of using an exponential decay function for the probability of an intervention step at time, t. For these simulations, the intervention urn contains one thousand red balls. This value is slightly lower than the 1,250 red balls that performed well in Figure 12, but this reduction is offset by increasing the number of nodes added at each intervention step. The initial Barabási-Albert graph was constructed with one connection added for each node, and each intervention node connects to 1,000 existing nodes in the network.

The first simulation varies the rate of exponential decay function, with the results shown in Figure 15. Based on the results from varying the intervention probability in Figure 13, the expected number of interventions can be estimated to lie between 360 and 720. To remain within this range, the simulations consider two intervention nodes per intervention step and use decay functions that produce a similar expected number of interventions.

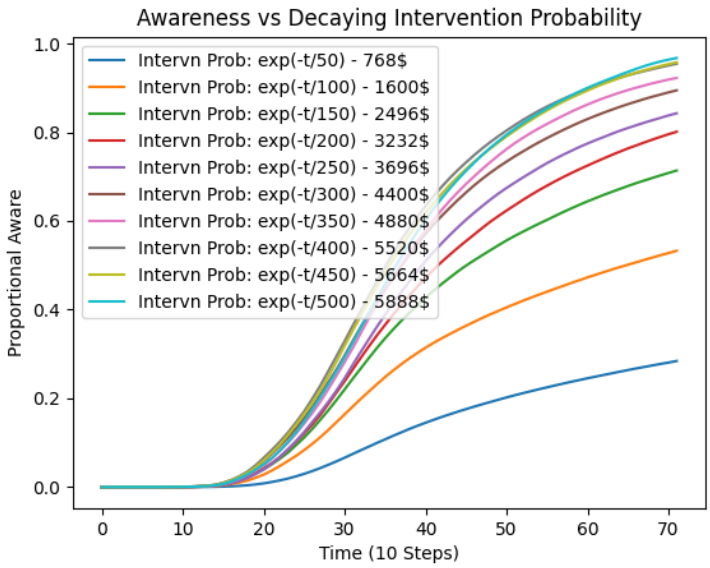


Figure 15: Awareness percentage for varying exponential decay intervention probability functions.

The results in Figure 15 show that faster decay rates lead to higher early intervention activity, resulting in a greater proportion of the network becoming aware and faster convergence to consensus. However, faster decay rates also lead to higher costs due to the increased number of early interventions. The results further indicate that beyond a certain point, specifically for decay functions slower than $e^{t/400}$, the improvement in final awareness is minimal, and all cases converge within similar time frames. Based on these findings, the decay function, $e^{t/200}$ was selected for further analysis, as it does not achieve full consensus with only two nodes and allows for clearer evaluation of additional interventions.

The second simulations examines the effect of varying the number of intervention nodes at each step while using the exponential decay function $e^{t/200}$. The results of this simulation are shown in Figure 16.

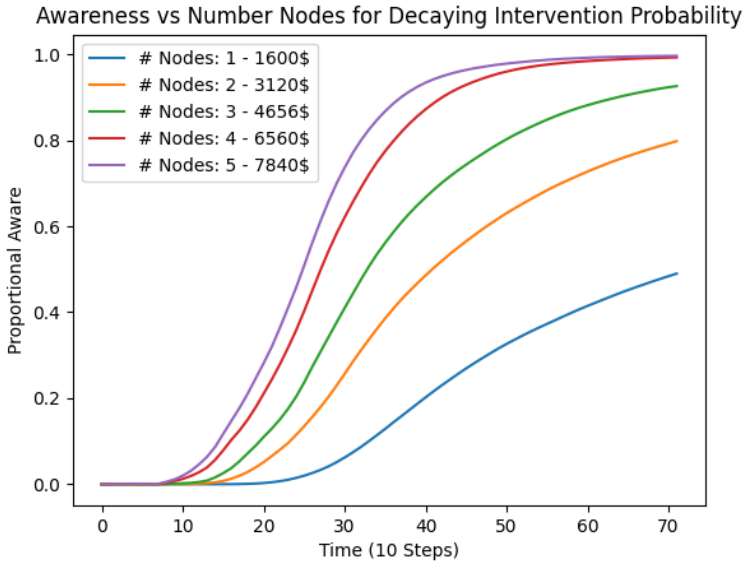


Figure 16: Awareness percentage for different intervention node counts with exponential decay probability.

Figure 16 follows expected trends: increasing the number of intervention nodes leads to a faster convergence, a higher propagation of aware nodes, and increased overall cost. Using four or five nodes per intervention step achieves full consensus, while three nodes approaches but does not reach complete consensus. The expected total number of interventions differs significantly across these cases, making direct comparison difficult.

6.1.6 Number of Intervention Nodes

To address the variation in the expected number of interventions, the next set of experiments select combinations of intervention probabilities and node counts that produce similar expected numbers of interventions in attempt to maximize the proportion of aware nodes. The results are presented in Figure 17.

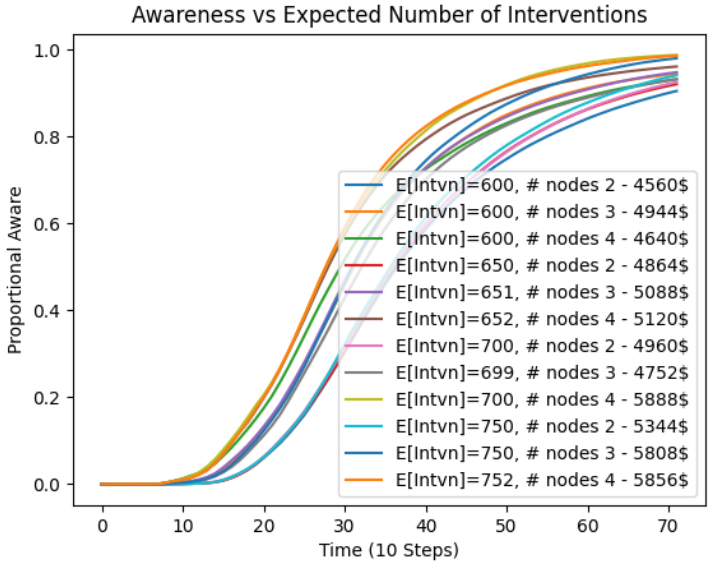


Figure 17: Awareness percentage for intervention methods with similar expected number of intervention.

From the results in Figure 17, changes in the expected number of interventions have only a small impact on the proportion of aware nodes. Simulations with more expected interventions achieve a slightly higher proportion of awareness, approximately 7-8% more, compared to those with fewer interventions, but at an additional cost of just over $1,000. Changes in the number of nodes per intervention step and the rate of decay have limited impact on the final level of awareness, but they do affect the timing of convergence. Specifically, scenarios with more nodes and faster decay begin converging earlier in the simulation. This earlier convergence is likely due to more nodes being introduced at the start, allowing it more time to propagate through the network. Based on these results, the final approach favours using more nodes per intervention step combined with a faster decay in intervention probability, that have an expected number of interventions around 750. When increasing the number of nodes in the graph, the expected number of injections will be increased by the same factor.

6.2 Model 2: Targeted Injection into Optimal Nodes: Simulation Results

The simulations for the direct injection intervention follow the setup described in Section 4.4.3 and Section 4.4.4 The costs for Model 2, the direct injection intervention, are calculated as explained in Section 4.5.

6.2.1 Number of Barabási-Albert Connections (m)

An initial simulation was conducted by varying the number of Barabási-Albert connections in the graph to examine the impact of network connectivity on the spread of awareness. To isolate this effect, the number of red balls injected into the network was fixed at 296, and the number of injections constant at 5,000. The

injection intervention was applied at step 144 to allow the Pólya process and memory to develop beforehand. The results are shown in Figure 18.

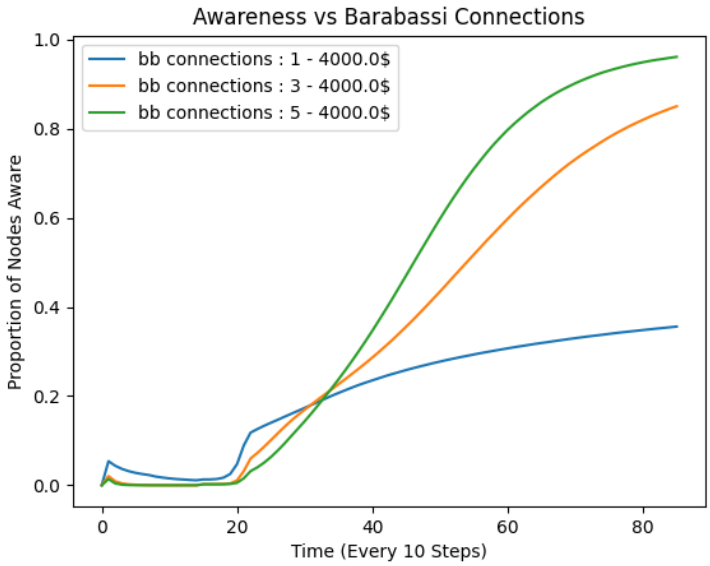


Figure 18: Awareness percentage for varying Barabási-Albert connection counts.

From the results in Figure 18, the case $m = 1$ shows minimal impact from the intervention, as awareness does not propagate through the network as effectively as in more connected graphs. This effect is likely due to the limited number of paths available for awareness to spread when propagating through the network. As a result, $m = 1$ Barabási-Albert networks are not considered further, since achieving consensus would require an impractically large amount of interventions to have the network to converge to a consensus. For $m = 5$, awareness propagates through the network the quickest, and reaches near full consensus. In the $m = 3$ case, the results indicate a trend towards full consensus, but the simulation ends before it is fully achieved. With additional parameter tuning, this network is likely to reach consensus. Going forward, $m = 3$ Barabási networks will be considered, as it provides a more balanced and informative case for studying convergence.

6.2.2 Number of Red Balls Per Injection

The next parameter evaluated was the number of red balls injected into each urn during an intervention step. For these simulations, the number of connections per each node in the Barabási-Albert graph is held constant at 3, and the number of injection nodes is fixed at 5,000. The results are shown in Figure 19.

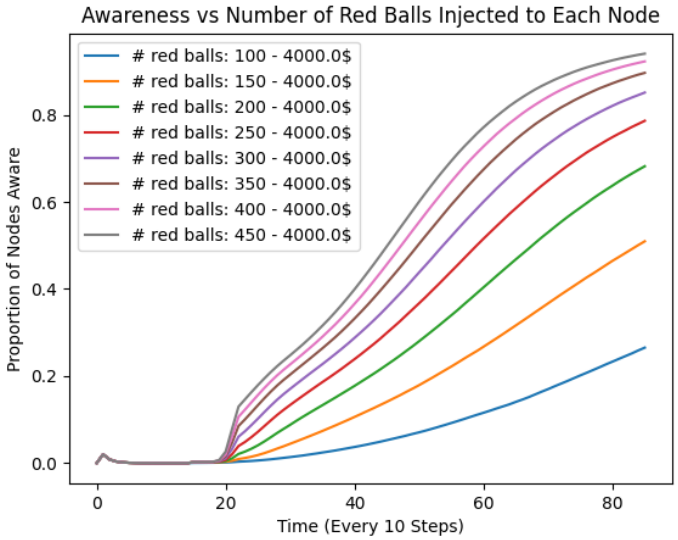


Figure 19: Awareness percentage in model for different amounts of red balls injected into the urns.

Figure 19 behaves as expected: as more balls are added to each injection node, the proportion of aware nodes increases. Across the simulations, once more than 300 red balls are added per injection, the increase in the final proportion of aware nodes becomes minimal. As additional balls are added, the gains in consensus diminish, indicating that increasing the number of red balls per injection is not the most effective way to drive the network towards consensus. Thus, the number of red balls added per injection will be fixed at 300 for future simulations.

6.2.3 Number of Injections

The final parameter evaluated was the number of nodes receiving injections. A $m = 3$ Barabási-Albert network was used, and the number of balls injected into each urn was constant at 300, shown in Figure 20.

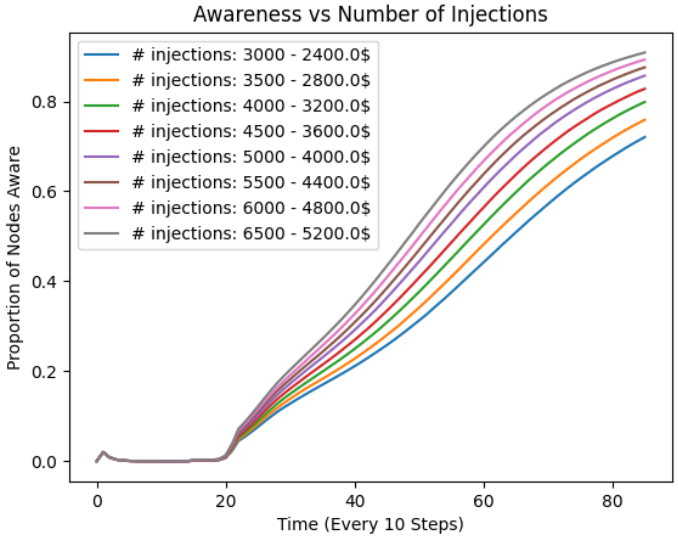


Figure 20: Awareness percentage in model for varying number of injections.

The simulations follow the expected trend where increasing the number of injections leads to a higher proportion of aware nodes at the end of the simulation. A similar pattern is observed when varying the number of red balls per injection (see Figure 19). However, the improvement in consensus becomes less significant as more nodes are added. This pattern suggests diminishing returns, where achieving full consensus would require a substantial increase in both injections and red balls, resulting in impractical costs. Overall, the results indicate that the number of connections in the network limits how efficiently awareness can spread through the graph for the direct injection approach.

7 Analysis

This section analyzes the simulation results to evaluate how different intervention strategies and model parameters influence information spread in the network Pólya process. Both direct injection and switch network interventions are compared across varying parameter configurations. The focus is on how key factors, including network density, intervention strength, frequency, and process memory, interact to affect convergence behavior and overall awareness in the network.

7.1 Parameter Interactions in Simulation

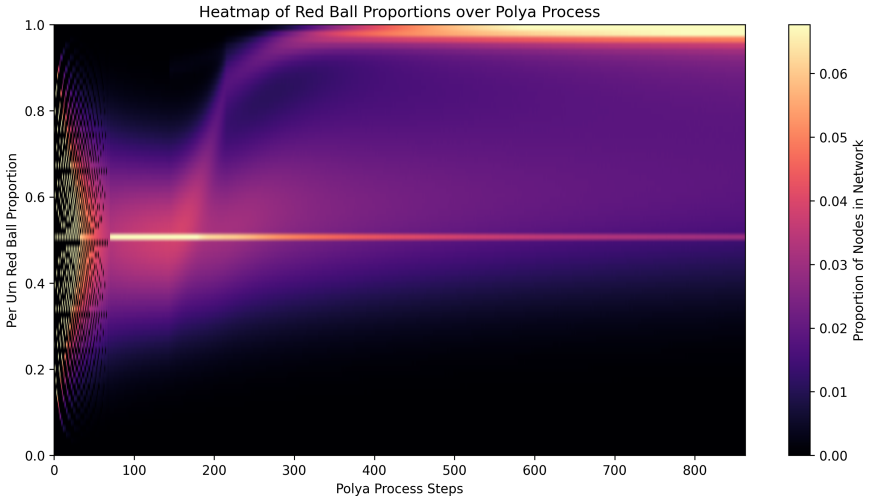
In the final full-scale simulations of the Pólya process, four parameters were selected for variation within each model. Table 7 outlines these parameters, while all simulated parameter combinations are detailed in Table 8 and Table 9, for the switch network and direct injection intervention, respectively. Each parameter exhibits complex interactions with the others, such that modifying a single parameter can lead to significant variation in simulation outcomes. The analysis combines both mathematical reasoning and empirical results, using heatmaps, histograms, and tabulated data to express the state of the network.

7.1.1 Barabási-Albert Interactions

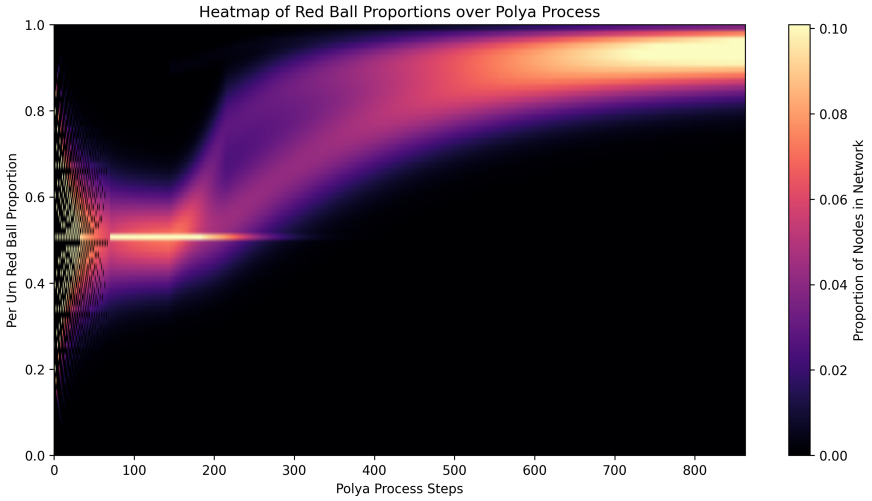
Increasing both the number of interventions performed and the number of red balls added per urn produces a compounding effect, raising the proportion of nodes with a strong red bias. However, the extent to which this bias spreads throughout the network is strongly influenced by the graph's density, as determined by the number of connections per node in the Barabási-Albert model.

During graph generation, if all newly added nodes in the Barabási-Albert network connect to only one or two existing nodes, the resulting graph is relatively sparse, leading to longer average shortest path lengths between nodes. In such low-density networks, information originating at a given node must pass through more intermediate nodes to reach the rest of the network compared to denser graphs. Since information propagation occurs through the Pólya process, which operates locally on first-order connections, a greater number of process steps is required for information to spread outward from the initial target nodes in a low-density graph compared to a high-density graph. This behaviour significantly slows the propagation of injected bias. Conversely, higher-density networks exhibit shorter path lengths, enabling faster dissemination of information and allowing the networked process to converge in fewer steps.

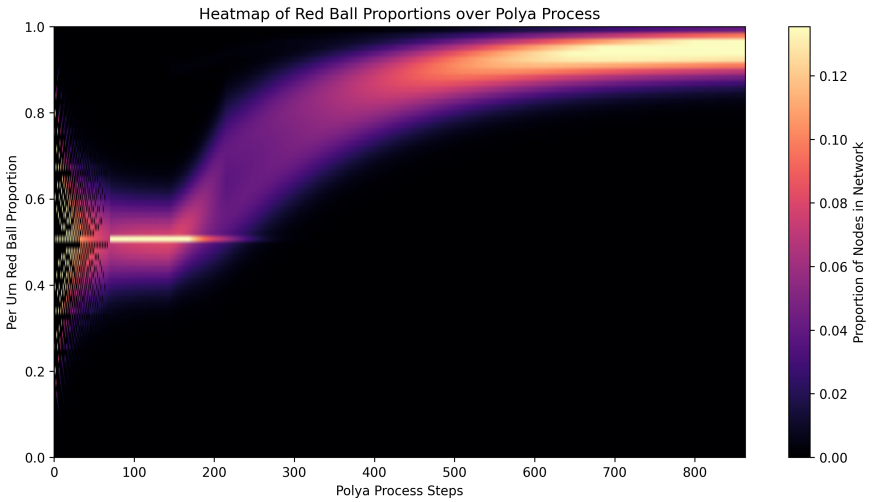
This relationship is illustrated through Figures 21a, 21b, and 21c. At the conclusion of the Pólya process, all three simulations exhibit a substantial proportion of nodes with strong bias. However, Figure 21a also shows a non-negligible number of nodes with intermediate red node proportion in the 0.5-0.8 range. This result indicates that while the direct injection intervention effectively influences nodes in close proximity to the targeted injection sites, the 864 step process duration is insufficient for the induced bias to fully propagate to more distant regions of the network.



(a) Simulation 7: Direct injection intervention on low-density graph.



(b) Simulation 16: Direct injection intervention on mid-density graph.



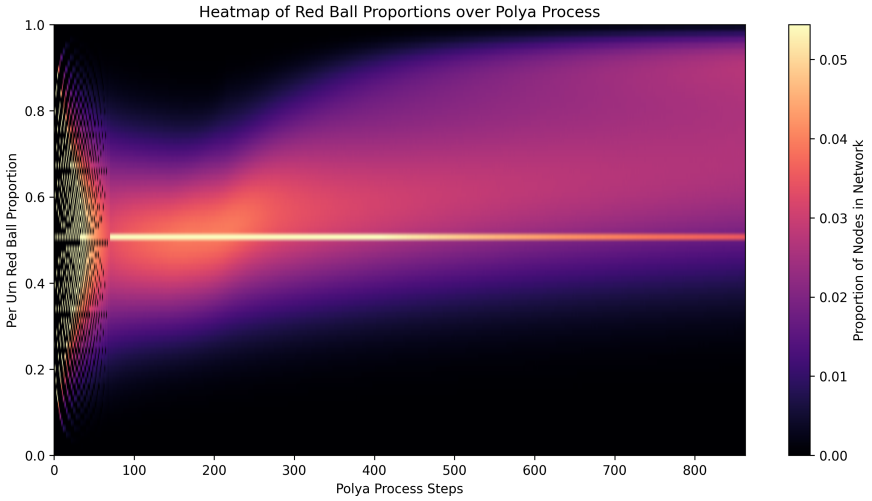
(c) Simulation 25: Direct injection intervention on high-density graph.

Figure 21: Change in convergence rate over Barabási-Albert per node connections 1, 3, and 5.

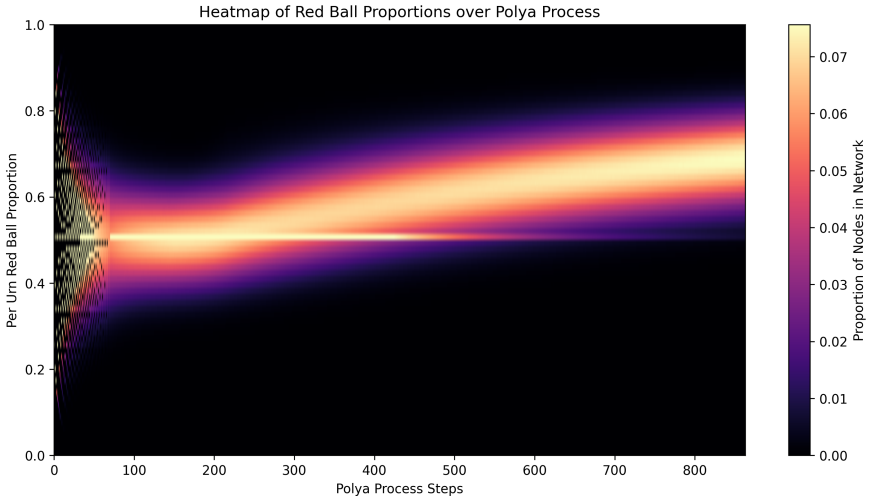
Simulations involving the switch network intervention, similar to those using direct injection intervention, demonstrate that information propagation is strongly influenced by the density of the underlying Barabási-Albert graph. However, unlike direct injection intervention, the switch network intervention actively alters the network by introducing new edges, thereby partially mitigating the limitations imposed by low-density connectivity.

In sparse networks, where information spreads slowly due to long average shortest path lengths, the switch network intervention can substantially enhance propagation by creating structural shortcuts. By connecting newly introduced red-biased nodes to existing high-degree nodes, the intervention reduces the effective distance between distant regions of the network and establishes additional pathways for the diffusion of red bias. Importantly, introducing multiple intervening nodes within a single intervention further amplifies this effect by creating several parallel routes for information.

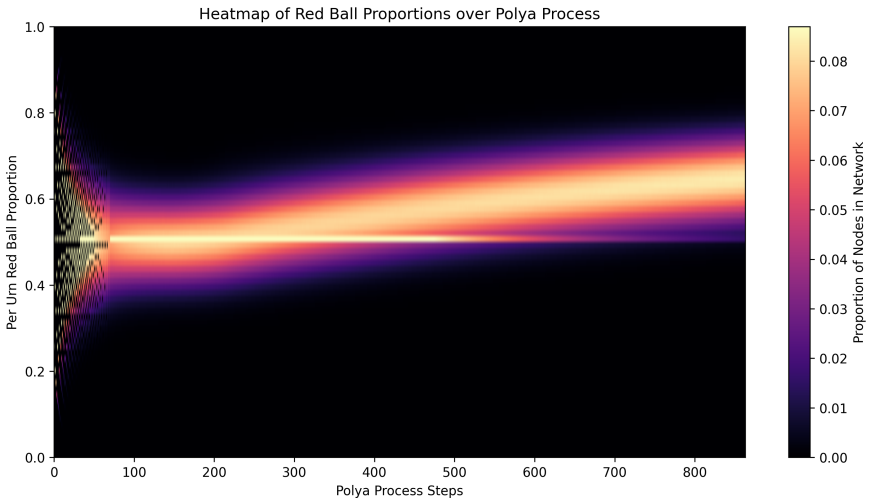
Nevertheless, the effectiveness of this strategy remains dependent on the underlying network density and the number of connections assigned to each added node. In highly sparse graphs, even when new nodes are attached to high-degree hubs, the total number of viable propagation pathways remains limited. Consequently, while the intervention establishes strong localized sources of red bias, the global diffusion of this bias continues to be constrained by the network's structural properties. These effects are illustrated in Figures 22a, 22b, and 22c.



(a) Simulation 62: Switch network intervention on low-density graph.



(b) Simulation 71: Switch network intervention on mid-density graph.



(c) Simulation 80: Switch network intervention on high-density graph.

Figure 22: Change in convergence rate over Barabási-Albert per node connections 1, 3, and 5.

Another key observation from the simulations is that increasing the density of the Barabási–Albert graph raises the number of red balls required for an intervention to meaningfully influence the Pólya process. This effect arises from the construction of the super urn at each node, which aggregates the urns of its neighbours.

In the network Pólya process, each node draws from a super urn composed of its own urn combined with those of its neighbouring nodes. As the graph density increases, each node acquires more neighbours, leading to larger super urns. While this facilitates faster information diffusion across the network, it also dilutes the impact of any individual intervention.

To illustrate this effect, consider a node with degree d. The total number of balls in its super urn is given by the sum of the urn sizes of its neighbouring nodes together with its own urn, and therefore increases with d. When red balls are introduced into a node's urn through an intervention, their effect is confined to the super urns of that node and its immediate neighbours. If these nodes have many neighbours, their individual urns constitute only a small fraction of the total balls in the corresponding super urn. Consequently, the probability of drawing a red ball from the super urn increases only marginally unless a substantial number of red balls are introduced through intervention.

In contrast, in sparse networks where nodes have fewer neighbours, each individual urn contributes makes up a larger proportion of the corresponding super urn. As a result, even a small number of injected red balls can substantially alter the composition of the super urn and, in turn, have a stronger influence on the Pólya process. Consequently, interventions are more efficient in sparse networks, as each injected ball has a greater proportional impact on the draw probabilities.

This observation highlights a fundamental tradeoff associated with graph density. While increased density enhances the speed of information propagation by reducing path lengths and increasing the number of interaction pathways, it simultaneously reduces the relative influence of any single urn by expanding the average total size of the super urns. As a result, denser networks require larger interventions, with more red balls, to achieve comparable local changes in draw probabilities relative to sparse networks. This behavior is supported empirically in Figures 22b and 22c, which compare medium and high density graphs under identical intervention settings including the same number of red balls per added node and an equal number of connections in each switch network intervention. In the medium density case, a higher mean red bias is achieved across the network than in the high density case.

7.1.2 Comparison of Degree and Eigenvector Centrality Direct Injection Interventions

Two direct injection intervention strategies were evaluated: targeting nodes with high degree centrality and targeting nodes with high eigenvector centrality. Across all tested Barabási-Albert graph densities, degree centrality intervention consistently outperformed eigenvector centrality intervention, in driving convergence of the networked Pólya process. This result can be understood by considering how the two centrality measures interact with both the Barabási-Albert graph structure and the super urn process.

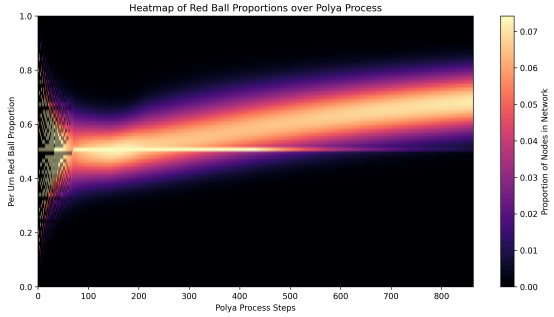
Degree centrality targets nodes with the greatest number of direct connections. The dynamics of the Pólya process imply that nodes with high degree centrality contribute their urns to many super urns at each step of the process. When red bias is injected into a high degree node, it is rapidly disseminated to many neighboring nodes, increasing the likelihood that these nodes will draw red balls and further reinforce red bias within their own urns. In a Barabási-Albert graph, this effect is further amplified by the densely connected core that emerges at the center of the network. In simulations, nodes within this core are typically the first selected for degree centrality based interventions. Since the neighbours of these target nodes are themselves highly connected members of the core, the Pólya process facilitates rapid diffusion of information throughout the clique and into the broader network.

Eigenvector centrality, by contrast, prioritizes nodes that are connected to other highly connected nodes, rather than those with the largest number of direct neighbours. While this captures global influence, it does

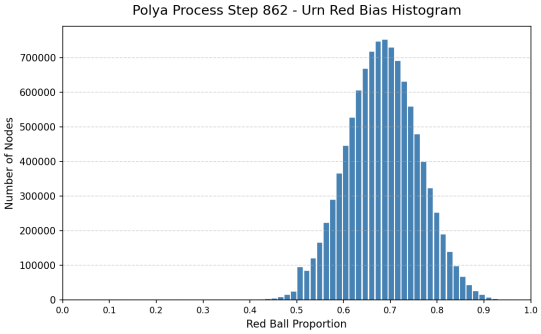
not guarantee high local connectivity. As a result, interventions targeting high eigenvector centrality nodes initially influence fewer super urns, leading to a slower and more indirect propagation of red bias through the network, which occurs through intermediate high degree nodes.

These dynamics are reflected in Figures 23a, 23c, 23e, and 23g. The proportion of nodes with equal red and black bias declines more rapidly in 23a than in 23c. In particular, the final step proportions of equally biased nodes in Figures 23b and 23d are 0.0092 and 0.0484, respectively. Empirically, this corresponds to 101,312 nodes remaining unaffected under degree centrality based injection, compared to 510,784 nodes in the eigenvector centrality case. This disparity arises because degree centrality targets nodes with the highest number of direct connections, enabling faster propagation of red bias through the network relative to eigenvector centrality, and resulting in a substantially lower number of unaffected nodes by the end of the simulation.

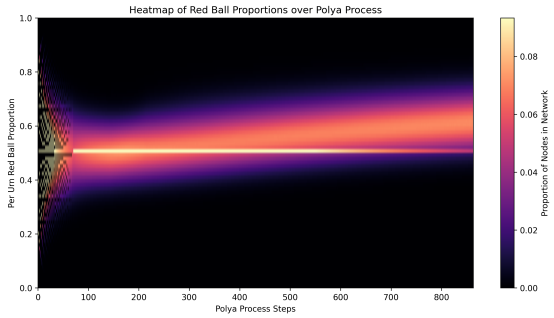
The results above also help explain the behaviour observed in Figures 23e and 23g. In both cases, two distinct trajectories emerge around step 150, indicating a delay in the propagation of red bias. The steeper trajectory corresponds to the initial spread from the injection sites to immediate neighbours, while the second, lagging trajectory reflects the propagation of red bias from those neighbours to the rest of the network. The system shown in Figure 23e converges to a high red bias more rapidly than that in Figure 23g, as evidenced by the steeper early slope of the lagging trajectory in Figure 23e. Although both injection strategies propagate bias to their immediate neighbours at the same rate, targeting highest degree nodes results in a larger initial set of influenced neighbors. Consequently, a greater proportion of nodes attain red bias earlier in the Pólya process, which then cascades through the network. This cascading effect explains why the system in Figure 23e converges more quickly, and reaches a stronger final red bias than that in Figure 23g.



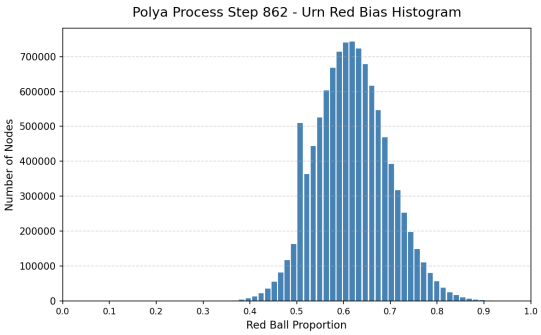
(a) Simulation 9: Degree centrality optimized direct injection intervention of 74 red balls on medium-density graph.



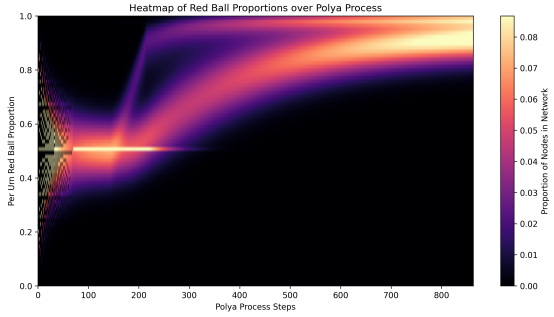
(b) Histogram of final Pólya process step in 23a.



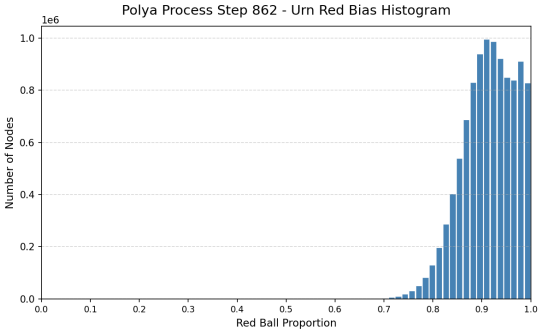
(c) Simulation 36: Eigenvector centrality optimized direct injection intervention of 74 red balls on medium-density graph.



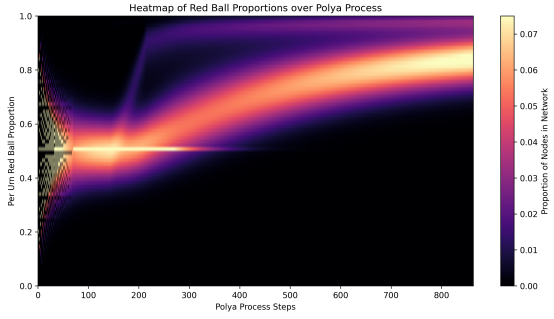
(d) Histogram of final Pólya process step in 23c.



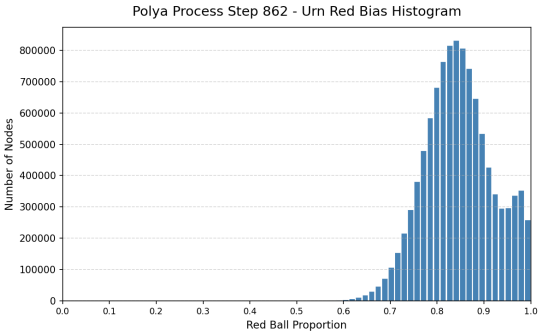
(e) Simulation 11: Degree centrality optimized direct injection intervention of 1,184 red balls on medium-density graph.



(f) Histogram of final Pólya process step in 23e.



(g) Simulation 38: Eigenvector centrality optimized direct injection intervention of 1,184 red balls on medium-density graph.



(h) Histogram of final Pólya process step in 23g.

Figure 23: Comparison of degree and eigenvector centrality optimized injection interventions over identical networks.

7.1.3 Memory Interactions

The finite memory of the Pólya process interacts with other simulation parameters in non-intuitive ways. Its most significant effect on the simulation is the constraint that it imposes on the total number of balls that can be present in any given urn at any step in the simulation. In particular, the node awareness is inversely proportional to the length of the Pólya process memory, as shown in the following equation, where $P_{N,K}(R)$ denotes the proportion of red balls at node N immediately after Pólya process step K, M is the length of the Pólya process memory, and $R_{initial}$ and $B_{initial}$ are the initial number of red and black balls in the urn.

$$P_{N,K}(R) = \frac{\sum_{i=max(1,\ K-M)}^{K} (\mathbb{1}_{\{\Delta_R \text{ Red added on step } i\}}) + R_{initial}}{M + (R_{initial} + B_{initial})} \tag{9}$$

The memory constraint limits the possible numbers and combinations of red and black balls that can exist in any urn at any step in the Pólya process after the memory limit is reached. To illustrate how this affects the process's ability to converge to a sufficiently high red bias, consider a Pólya process initialized with one red and one black ball, $\Delta_R = 1$, a memory of two, and no removal of initial balls. Under these conditions, any nodes in this process's graph is restricted to the following possible ball ($\{R, B\}$) combinations: $\{R, B\} = \{\{1, 1\}, \{1, 2\}, \{2, 1\}, \{2, 2\}, \{3, 1\}, \{1, 3\}\}$. Consequently, the maximum achievable red bias is $\frac{3}{4}$, meaning that no node in the network can ever become fully aware of the food recall under these constraints.

For this reason, the Pólya process memory length must be sufficiently large to allow for complex behaviour in the networked process. However, as memory increases, the influence of each individual Pólya process step on overall red convergence decreases proportionally and independently of Δ_R. This implies that interventions performed on the graph before M steps of the Pólya process have a substantially greater effect than those performed at M steps and beyond the memory threshold. Additionally, the number of red balls introduced during intervention must be scaled appropriately with respect to both M and the number of Barabási-Albert connections in the network in order to maintain comparable effects across configurations.

In the final simulations, the Pólya process memory was set to 72, meaning that after step 72, each node's urn size remains fixed at 74 for the remainder of the simulation. As a result, each node can only occupy 72 distinct post-memory states, of which 14 correspond to "aware" states under the chosen 80% awareness threshold.

7.1.4 Comparison of Direct Injection and Switch Network Intervention Strategies

The direct injection and switch network interventions present two different approaches to influencing the convergence of the network Pólya process. The direct injection intervention interacts with the process by modifying the urn compositions of existing nodes, whereas the switch network intervention modifies both the urn compositions and the network's structure by introducing new nodes and edges. To compare these approaches rigorously, it is useful to analyze their effects in terms of their contribution to the super urns that govern the Pólya process.

In the direct injection intervention, a total of $T = r \times k$ red balls are distributed across k nodes, with r balls added to each node. The effect of this intervention on the network depends on how these injected balls influence the super urns of neighboring nodes. For a node i, the probability that a red ball will be drawn from it's super urn and added to it's urn is given by:

$$P_i(\text{red}) = \frac{\sum_{j \in \mathcal{N}(i) \cup \{i\}} R_j}{\sum_{j \in \mathcal{N}(i) \cup \{i\}} (R_j + B_j)} \tag{10}$$

where R_j and B_j denote the number of red and black balls in node j's urn. Injecting red balls increases R_j instantly for injected nodes and rapidly for their direct neighbours, but does not change the structure of the

immediate neighbourhood $\mathcal{N}(i)$.

In contrast, each new node added through the switch network intervention contributes both additional red balls and additional edges to the network. If a new node v is connected to a set of nodes S, then for each node $i \in S$, the super urn is modified by the addition of the new node's urn:

$$P_i(\text{red}) = \frac{R_v + \sum_{j \in \mathcal{N}(i)} R_j}{(R_v + B_v) + \sum_{j \in \mathcal{N}(i)} (R_j + B_j)} \tag{11}$$

This mechanism directly increases the red proportion in the super urns of all nodes in S, influencing them towards a red bias which will in time influence their immediate neighbours towards a red bias.

To compare the effectiveness of the two interventions, consider a scenario in which the one hundred most degree central nodes in the network are injected with enough red balls so that their super urns almost always select a red ball. In this case, these 100 nodes and all of their immediate neighbours, rapidly become fully red biased, with only red balls remaining. In contrast, a single switch network intervention connecting to these one hundred degree central nodes produces a similar effect, as the super urn's of these central nodes incorporate the added node and quickly converge towards a red bias. However, a key distinction emerges: the immediate neighbours of the central nodes experience a delayed influence under the switch network intervention, since their super urns do not directly include the new node's urn in their super urns.

When cost is not a constraint, direct injection is more effective for rapidly spreading bias, as it can immediately influence targeted nodes and quickly propagate awareness to their neighbours. However, the switch network intervention achieves a similar initial effect with a single intervention, biasing the same targeted nodes within the time it would take for direct injection to reach neighbouring nodes. Moreover, multiple switch network interventions can interconnect and extend to neighbouring nodes of those central nodes, enabling a wider spread of red bias in fewer steps compared to direct injection. This behavior is consistent with the findings in Section 7, suggesting that switch network interventions with high connectivity outperform direct injection when both the number and strength of interventions are constrained by cost.

7.2 Results

Using the WEM framework, both intervention strategies were evaluated based on the metrics listed in Table 3. Scores range from 1 to 5, where 5 indicates strong performance and 1 indicates weak performance, as defined in Table 6 of the Appendix.

Metric	**Weighting (1-5)**	**Model 1 Score (1-5)**	**Model 2 Score (1-5)**
Economic	4	5	3
Environmental	2	3	3
Social	3	4	3
Feasibility	3	4	3
Safety	5	4	4
Complexity	3	5	2
Scalability	2	4	2
Ability to reach consensus	5	4	4
Propagation speed	3	4	5
Total Weighted Score		125	101

Table 3: Weighted evaluation matrix comparing intervention strategies across key criteria, with scores ranging from 1 (low performance) to 5 (high performance).

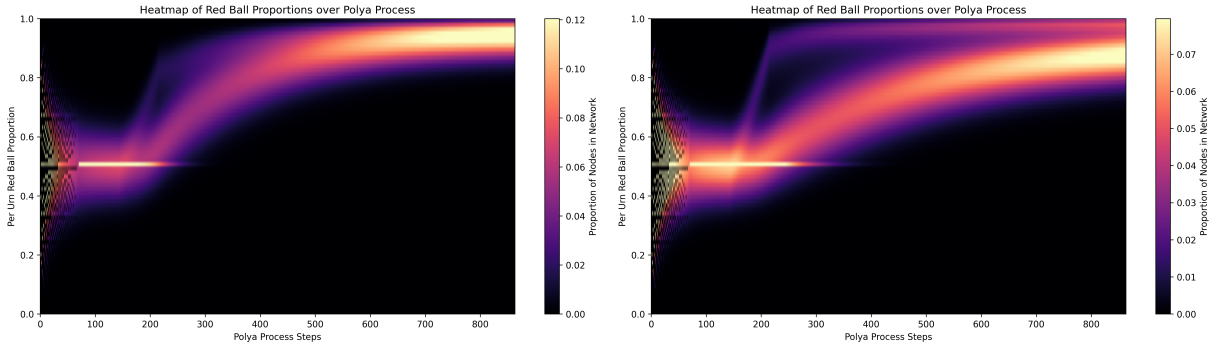
Based on both the simulation results and the weighted evaluation matrix, the switch network intervention, Model 1, consistently outperforms direct injection, Model 2, particularly in terms of cost efficiency. Direct

injection requires either direct messaging or paid social media advertising, with costs typically ranging from \$5 to \$15 per thousand impressions [21]. These costs increase further when targeting highly central users, who often charge a premium for access to their audiences.

In contrast, the switch network intervention leverages existing, highly followed government social media accounts to disseminate information. Once information is shared through these well-connected accounts, it can be amplified by journalists and news organizations, facilitating broader organic diffusion across the network. This approach results in a substantially lower cost per impression compared to paid advertising and scales more efficiently in large networks.

Specifically, as analyzed in Section 4.5, when costs are scaled to reflect real world conditions, the switch network intervention incurs an approximate cost of \$6,000 per recall, compared to \$8,000 per recall for direct injection. Compared to the Canadian Food Inspection Agency's estimated \$8,400 budget per Class I food recall, the switch network approach remains feasible and allows for multiple interventions within a single intervention [6] [7]. In contrast, direct injection nearly exhausts the entire campaign budget with a single intervention, significantly limiting its practical applicability. A detailed breakdown of the cost calculation for the Canadian Food Inspection Agency's Class I recall budget is provided in Section 12.5 of the Appendix.

The simulation results further support this conclusion, demonstrating that the switch network intervention, Model 1, achieves faster and more widespread convergence toward the red state. Figure 24, Table 4, and Table 5 compare a switch network intervention and a direct injection intervention simulation under equal costs, demonstrating the switch network's improved ability to reach a red bias under these fixed cost and time constraints.



(a) Fixed \$6,000 cost simulation with switch network intervention.

(b) Fixed \$8,000 cost simulation with direct injection intervention.

Figure 24: Final, fixed cost simulations for switch network and injection interventions over 10 million node graph.

Red Ball Proportion	Number of Nodes
$0.0 - 0.1$	0
$0.1 - 0.2$	0
$0.2 - 0.3$	0
$0.3 - 0.4$	0
$0.4 - 0.5$	0
$0.5 - 0.6$	0
$0.6 - 0.7$	600
$0.7 - 0.8$	$79,961$
$0.8 - 0.9$	$2,603,529$
$0.9 - 1.0$	$7,869,319$
Number of aware nodes $(0.8 - 1.0)$	$10,472,848$
Proportion of aware nodes	0.9924

Table 4: Awareness distribution for Pólya process step 862 in switch network intervention simulation in Figure 24.

Red Ball Proportion	Number of Nodes
$0.0 - 0.1$	0
$0.1 - 0.2$	0
$0.2 - 0.3$	0
$0.3 - 0.4$	0
$0.4 - 0.5$	0
$0.5 - 0.6$	914
$0.6 - 0.7$	$59,885$
$0.7 - 0.8$	$1,257,616$
$0.8 - 0.9$	$5,357,550$
$0.9 - 1.0$	$3,877,443$
Number of aware nodes $(0.8 - 1.0)$	$9,234,993$
Proportion of aware nodes	0.8751

Table 5: Awareness distribution for Pólya process step 862 in direct injection intervention simulation in Figure 24.

Table 4 and Table 5 indicate that awareness is strongly concentrated in the highest red ball proportion ranges for both intervention strategies at step 862. The switch network simulation shows that the vast majority of nodes fall within the 0.8–1.0 range, resulting in an overall awareness proportion of 99.24%, which suggests near complete diffusion across the network. The direct injection approach demonstrates a high level of awareness as well, but it yields a lower overall proportion of 87.51% and a more distributed spread across intermediate ranges. These findings further indicate that, while both interventions are effective and achieve consensus, the switch network approach leads to greater global consensus and a more uniform, widespread adoption of awareness throughout the system.

8 Impact of Engineering

The developed model for each strategy incorporates the social, economic, and environmental considerations of the Triple Bottom Line. The final model aims to optimize each objective while evaluating the effectiveness of the spread of awareness. Each component is accounted for to ensure a balanced and comprehensive outcome.

The results determined from the model iterations enable institutional stakeholders, such as government agencies, to develop an understanding of how to most effectively communicate and reinforce recall messages to the public. This understanding will help minimize the risks posed by recalled goods, thereby improving public health outcomes. Additionally, companies may be more inclined to prioritize transparent and timely

communication regarding affected products, fostering greater trust among consumers as well as at the governmental and institutional levels. A key social consideration of this model is the accessibility of information for marginalized communities, as barriers in access may limit the overall reach and effectiveness of recall awareness efforts.

The modelling analysis presents several economic benefits when applied to existing communication networks. With an improved understanding of how information propagates through social media, organizations can determine how to efficiently deliver critical information without incurring unnecessary costs. Increasing dissemination efficiency reduces resource expenditure for companies issuing recalls and institutions responsible for public safety communication. As shown in Section 7.2, implementing the strategy used in the switch network model achieves sufficient public awareness at the cost of $6,000, representing a 29% decrease from the estimated current budget of $8,400 allocated to each Class I recall in Canada [6] [7].

From an environmental perspective, the results of this analysis demonstrate potential reductions in waste and resource consumption. Improved efficiency in food recall awareness enables more precise identification and removal of affected products, reducing unnecessary disposal of uncontaminated food, and minimizing landfill use and greenhouse gas emissions. More effective dissemination also supports faster response times across the supply chain, limiting downstream waste at the retail and consumer levels. Furthermore, reducing redundant or ineffective communications decreases the consumption of physical materials, and limits the energy associated with data storage and information distribution.

9 Ethics, Equity, and Standards

During model development, economic impact and efficiency of solutions were critical parameters in determining model success. However, prioritization of ethical and equitable solutions is crucial in ensuring the due dillegence to public health and safety is met. The Professional Engineers Ontario (PEO) Code of Ethics and Canadian Government standards were considered throughout the design process and shaped the framework of the project goal and chosen advertisement strategies.

In particular, Section 1.ii of the PEO Code of Ethics, the duty to act at all times with fidelity to public is considered applicable as the project aims to increase the spread of awareness and thus serves the public need for communication of health and safety risks [22]. Additionally, Section 2.i of the PEO Code of Ethics, the need to regard the engineer's duty to public welfare as paramount is applicable in that public safety and timely notification should be prioritized above the cost of advertisement campaigns [22].

Under the Canadian Food Inspection Agency Act, the CFIA is responsible for food safety investigations and issuing food recalls when deemed a risk to public, animal, or plant health [23]. When a food safety investigation is triggers, the CFIA is responsible for conducting a risk assessment, and assigning it into one of three categories; Class I, Class II, and Class III [24] A Class I recall involves a high risk food item which may cause serious health problems or death [24]. A Class II recall is considered moderate risk, and may lead to short-term or non-life threatenening health problems [24]. A Class III recall involves low risk, undesirable health problems, and includes food that does not comply with legislation [24]. Class I recalls were considered most crucial and require timely, effective, and widespread notification.

The PEO Code of Ethics and CFIA Act principles highlighted above were implemented into project goals, evaluation metrics, and simulation parameters. The project goal is to maximize awareness of food recall campaigns while achieving a cost threshold, rather than minimizing costs for a fixed percentage of awareness. In Section 4.7, the highest weighted metric was safety and ability of the model to reach consensus. Project design parameters including frequency of interventions, duration of advertisement campaigns, and percentage of consensus are aligned with that of Class I recalls. Additionally, other ethical and equitable considerations were considered in the model design. Targeted ads are chosen based on the number of connections a user has, as well as widespread dispersion in the network to avoid high awareness in one subgroup, but low awareness in another. Intervention strategies are not based on external factors of a user such as location of residents, occupation, age, or spoken language. Additionally, it was considered that not all individuals in the popula-

tion have social media, and a portion of available budget goes to alternative advertising strategies such as traditional news outlets.

10 Recommendations

Based on the simulation results, the switch network intervention, Model 1, is recommended as the preferred intervention strategy. Compared to direct injection, Model 2, this approach achieves high levels of awareness while requiring fewer resources, making it more suitable under the current Canadian budget constraints associated with food recall campaigns.

A key finding of this project is that the timing of interventions plays a critical role in determining each interventions effectiveness. Interventions applied early in the process have a greater impact on awareness propagation than those applied later. As a result, intervention strategies should prioritize high activity immediately following a recall, followed by a gradual reduction over time. Modeling the probability of intervention using an exponentially decreasing function provides an effective way to achieve this result, as it concentrates resources when they are most impactful while reducing unnecessary cost once awareness begins to spread naturally.

The simulations also show that several parameters exhibit diminishing returns. Increasing the number of red balls per intervention, the number of intervention nodes, or the probability of intervention beyond moderate levels leads to only small improvements in awareness while significantly increasing cost. This result indicates that efficient strategies should focus on maintaining moderate parameter values and distributing interventions across time, rather than concentrating resources into large individual actions.

The structure of the underlying network is another important factor that must be considered. In networks with low connectivity, awareness spreads slowly and may require additional intervention to reach a large portion of the population. In more highly connected networks, awareness spreads more quickly, but individual interventions have a reduced impact due to the larger number of connections influencing each node. As a result, intervention strategies should be adapted to the expected structure of the network to achieve the best results.

When direct injection is used, targeting nodes with high degree centrality is the most effective approach. These nodes influence a larger number of neighbours directly, allowing awareness to spread more quickly through the network. Targeting nodes based on eigenvector centrality was found to be less effective, as it does not maximize immediate exposure.

10.1 Future Work Recommendations

In addition to these practical recommendations, several improvements to the model are suggested to enhance its realism and applicability. First, allowing the network to grow dynamically during simulations would provide a more accurate representation of real-world social networks. In practice, platforms such as Instagram are not static; new users join continuously, and new connections are formed over time. Incorporating dynamic network growth into the Pólya framework would allow the model to capture how awareness spreads in an evolving system, and would make it possible to study how the timing of interventions interacts with periods of rapid network expansion.

Further analysis of the reinforcement parameters, specifically, Δ_R and Δ_B would provide deeper insight into how information spreads under different behavioural assumptions. These parameters directly control how strongly past exposure influences future behaviour. Varying these values would allow the model to represent different levels of user engagement, such as highly responsive users who quickly adopt and share information, versus more passive users who require repeated exposure. Understanding how sensitive the system is to these parameters would help determine how robust the intervention strategies are across different user populations.

Finally, additional testing across a wider range of parameter combinations is necessary to improve the reliability of the results. While this study focuses on specific values for intervention rates, network sizes, and decay functions, real-world conditions can vary significantly. Exploring a broader set of scenarios, including larger networks, different decay rates, and varying levels of intervention intensity, would help identify whether the observed trends remain consistent. This type of analysis would also make it possible to determine optimal parameter ranges under different conditions, improving the model's applicability across a wider range of scenarios and its usefulness for practical applications.

11 Conclusions

This thesis examined how food recall information spreads throughout networks using a Pólya process framework, with the objective of identifying strategies that maximize awareness while minimizing cost. Two intervention methods were analyzed: switch network (Model 1) and direct injection (Model 2), each representing a different approach to disseminating information in a network environment.

Within each model, multiple strategies were explored to evaluate how different parameter choices influence awareness propagation. For the switch network intervention, variations included the number of connections per node in the Barabási–Albert network, the number of intervention nodes introduced at each step, the number of red balls assigned to each intervention node, and the probability and timing of interventions. In particular, the introduction of an exponentially decaying intervention probability was shown to improve efficiency by concentrating effort at the beginning of the simulation, when it has the greatest impact. For the direct injection intervention, strategies focused on the selection of target nodes, including degree centrality and eigenvector centrality approaches, as well as the number of injections and the number of red balls introduced per injection. Across these variations, degree centrality consistently outperformed eigenvector centrality, as it allowed for more immediate and widespread influence through highly connected nodes.

From a cost perspective, the direct injection intervention follows a cost-per-thousand impressions model, averaging approximately $8 per 1,000 users [18]. As a result, the total cost scales proportionally with both the number of users targeted and the number of interventions performed. In large-scale simulations, a single injection targeting 10,000 users corresponds to a total cost of approximately $8,000 per recall, which is comparable to the estimated $8,400 available per Class I food recall [6] [7]. This comparison suggests that reliance on direct injection alone would constrain the number of interventions feasible within a realistic campaign budget. In contrast, the switch network intervention incurs a lower overall cost of approximately $6,000 per recall, which allows a greater number of interventions within the same budget.

In addition to cost, the effectiveness of each intervention strategy also depends on how they interact with the structure and dynamics of the network. The switch network intervention consistently produces strong and sustained propagation by leveraging and amplifying the underlying network structure, introducing highly connected nodes that enable awareness to spread more organically. This approach supports repeated, strategically timed interventions that reinforce awareness early and improve overall resource efficiency. In contrast, the direct injection intervention relies on influencing selected nodes, resulting in rapid but localized increases in awareness that require repeated application to achieve broader impact.

Beyond these differences in intervention design, several structural and dynamical factors further influence propagation outcomes. Network connectivity helps determine both the speed and efficiency of information spread. In denser networks, awareness propagates more quickly due to shorter path lengths and a greater number of transmission pathways. However, this increased connectivity also dilutes the effect of individual interventions, as each node's super urn incorporates a larger number of neighbouring urns, reducing the relative impact of injected bias. As a result, more substantial interventions are required to achieve the same local effect in highly connected networks.

The timing of interventions is equally critical. The simulations consistently show that early interventions have a disproportionately large impact on overall outcomes due to the reinforcement mechanism of the Pólya process. Initial increases in awareness were amplified over time, allowing early-stage interventions to shape the

trajectory of the entire system. Once awareness begins to propagate, it becomes increasingly self-sustaining, reducing the marginal benefit of later interventions. This result supports strategies that concentrate resources early, such as the use of exponentially decaying intervention probabilities.

In addition, many parameters exhibit diminishing returns. Increasing the number of red balls per intervention, the number of intervention nodes, or the probability of intervention initially improves both the speed and extent of awareness propagation. However, beyond moderate thresholds, further increases produce only marginal gains while significantly increasing cost. These results indicate that optimal strategies lie within a balanced range of parameter values, where sufficient influence is introduced to initiate propagation without unnecessary expenditure.

Overall, the findings of this project demonstrate that effective food recall communication requires strategies that align with both economic constraints and network dynamics. While direct injection can provide short-term increases in awareness, its cost structure limits its practicality at scale. In contrast, the switch network intervention offers a balanced and scalable approach, combining cost-efficiency with strong propagation dynamics to achieve widespread and sustained awareness. Therefore, the switch network intervention emerged as the optimal intervention strategy within the scope of this thesis as it achieves a strong balance between cost efficiency, scalability, and effective propagation dynamics. By prioritizing early intervention, leveraging network structure, and carefully selecting parameter values, organizations can improve the effectiveness of food recall communication while remaining within realistic budget constraints.

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12 Appendix

12.1 Github Code

The code for this project can be found using the URL below:
`https://github.com/Douglass-Jeffrey/MTHE493_Capstone_Polya_Process_with_Switched_Networks`

12.2 Evaluation Metrics

Metric	**Low Rating (1)**	**High Rating (5)**
Economic	The model's projected cost is significantly over the existing advertising budget for food recall campaigns.	The model's projected cost is at least 10% lower than the current budget for food recall campaigns.
Environmental	Implementation of the model would result in negative environmental impacts, characterized by high material and energy consumption and significantly elevated levels of carbon emissions and waste generation.	Implementation of the model is associated with minimal environmental impact, characterized by negligible material and energy consumption and minimal increases in carbon emissions and waste generation.
Social	The model has severe contrast in distributions throughout the network. Many subgroups do not reach consensus.	The model has relatively uniform distribution through the network. No unaware subgroups exist.
Feasibility	Implementation of the model is not possible in existing media advertisement infrastructure.	Implementation of the model is feasible in current media advertisement infrastructure. Integration would be efficient.
Safety	The model does not prioritize the public health and safety of individuals. Low levels of consensus are reached, propagation speed is slow, and many unaware subgroups exist within the network.	The model effectively prioritizes public health and safety by recognizing the urgency of food recall campaigns. High levels of consensus are achieved, propagation speed is rapid, and awareness is uniformly distributed across the network.
Complexity	Model simulation is extremely difficult and consumes a high level of storage/computational capacity and/or takes an extremely high simulation time.	Model simulation requires low levels of storage/computational capacity and takes a low level of simulation time.
Scalability	The model only considers a small percentage ($< 1\%$) of the Canadian population. Scaling of the results would not be considered accurate.	The model considers a relatively large percentage ($> 30\%$) of the Canadian population. The model is representative of social media networks.
Ability to reach consensus	Final consensus levels fall between 0% to 20%.	Final consensus levels fall between 80% to 100%.
Propagation Speed	The model cannot reach consensus by the end of the simulation.	The model reaches consensus by the end of the simulation. The model reaches 80% consensus by halfway through the simulation.

Table 6: Description of low level (1) and high level (5) ratings for each evaluation metric.

12.3 Summary of Simulation Parameters

Parameter	**Parameter Definition**	**Model 1**	**Model 2**
Number of Barabási-Albert connections (m)	The number of edges each new node forms with existing nodes during initial BA network generation. Higher values correspond to denser networks.	✓	✓
Number of red balls per intervening node	The number of red balls in the intervening node at the time of its addition.	✓	
Probability of intervention	The probability that a red ball is drawn from the super urn.	✓	
Number of connections	The number of edges created between the intervening node and existing nodes in the network.	✓	
Number of red balls per injection	The number of red balls directly added into each selected nodes.		✓
Number of injections	The number pre-existing nodes that will receive a fixed, memoryless addition of red balls to their urn.		✓

Table 7: Varied simulation parameters for switch network, Model 1, intervention and direct injection, Model 2, intervention.

12.4 Test Parameters

Table 8 shows the parameter sets used in each test conducted with varying Barabási connections (1, 3, and 5), using the switch network, Model 1, intervention method. The number of injections was varied based on the probability of intervention and the number of nodes added at each intervention.

Test	Initial Nodes	Intervention Method	Intervention Urn - Number of Red Balls	Pólya Steps	Memory Decay Time	Delta Gain
1	10,553,408	Centrality	1000	720	72	1
2	10,553,408	Centrality	1000	720	72	1
3	10,553,408	Centrality	1000	720	72	1
4	10,553,408	Centrality	1000	720	72	1
5	10,553,408	Centrality	1000	720	72	1
6	10,553,408	Centrality	1000	720	72	1
7	10,553,408	Centrality	1000	720	72	1
8	10,553,408	Centrality	1000	720	72	1
9	10,553,408	Centrality	1000	720	72	1

Table 8: Parameter sets used for tests with varying Barabási connections (1, 3, and 5) under the switch network intervention method.

Table 9 shows the parameter sets used in each test conducted with varying Barabási connections (1, 3, and 5), using the direct injection, Model 2, intervention method.

Test	Initial Nodes	Intervention Method	Intervention Steps	Number of Injections	Intervention Urn - Number of Red Balls	Pólya Steps	Memory Decay Time	Delta Gain
1	10,553,408	Centrality	144	16,384	74	864	72	1
2	10,553,408	Eigenvector	144	16,384	74	864	72	1
3	10,553,408	Centrality	144	16,384	296	864	72	1
4	10,553,408	Eigenvector	144	16,384	296	864	72	1
5	10,553,408	Centrality	144	16,384	1,184	864	72	1
6	10,553,408	Eigenvector	144	16,384	1,184	864	72	1
7	10,553,408	Centrality	144	32,768	74	864	72	1
8	10,553,408	Eigenvector	144	32,768	74	864	72	1
9	10,553,408	Centrality	144	32,768	296	864	72	1
10	10,553,408	Eigenvector	144	32,768	296	864	72	1
11	10,553,408	Centrality	144	32,768	1,184	864	72	1
12	10,553,408	Eigenvector	144	32,768	1,184	864	72	1
13	10,553,408	Centrality	144	65,536	74	864	72	1
14	10,553,408	Eigenvector	144	65,536	74	864	72	1
15	10,553,408	Centrality	144	65,536	296	864	72	1
16	10,553,408	Eigenvector	144	65,536	296	864	72	1
17	10,553,408	Centrality	144	65,536	1,184	864	72	1
18	10,553,408	Eigenvector	144	65,536	1,184	864	72	1

Table 9: Parameter sets used for tests with varying Barabási connections (1, 3, and 5) under the direct injection intervention method.

12.5 Cost Breakdown for the Current Canadian Food Recall Campaign Budget

To calculate the average cost spent by the CFIA per class I recall, the following equation was used:

$$C_{CFIA/recall} = C_{CFIA,a} \times N_{recall/a} \times \%_{digital} \times \%_{social\ media} \tag{12}$$

where

- $C_{CFIA/recall}$ is the average amount the CFIA spends per Class I recall.
- $C_{CFIA,a}$ is the CFIA annual spend on food recalls. In 2023, it was $\$1,487,962$ [6].
- $N_{recall/a}$ is the average number of food recalls per year, averaged over five fiscal years from 2020 to 2025, 92 [3].
- $\%_{digital}$ is the percentage that the Government of Canada spends on digital marketing as opposed to other methods, 63% [6].
- $\%_{social\ media}$ is the percentage of digital marketing on social media and display programmatics 82% [6].

The calculation can be seen below:
$C_{CFIA/recall} = C_{CFIA,a} \times N_{recall/a} \times \%_{digital} \times \%_{social\ media}$
$C_{CFIA/recall} = \$1,487,962 \times \frac{106+114+83+57+96}{5} \times 0.63 \times 0.82$
$C_{CFIA/recall} \simeq \$8400$